Metallurgy Kinetics

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Yildiz Technical University, Department of Metallurgical & Materials Engineering
Fall 2026-2027

METALLURGY KINETICS - MSE 4621
Fall Semester 2026 - 2027
Day / Time: Monday / 13:00 - 14:50

Week 1	28-09-2026	Introduction to kinetics, determination of the reaction rate, factors affecting reaction rates
Week 2	05-10-2026	Classification of reactions, kinetics of homogeneous reactions, degree of reaction, zero-order reactions
Week 3	12-10-2026	1st, 2nd, 3rd, n^{th} degrees reactions, problem solving
Week 4	19-10-2026	Method for determining reaction order, problem solving
Week 5	26-10-2026	Reversible, serial, parallel reactions
Week 6	02-11-2026	Effect of temperature on reaction rate
Week 7	09-11-2026	Effect of temperature on reaction rate
Week 8	16-11-2026	**Midterm**
Week 9	23-11-2026	Activation energy depending on temperature, Le Chatelier Principle
Week 10	30-12-2026	Theories of reaction kinetics, problem solving
Week 11	07-12-2026	Heterogeneous reactions, factors affecting reaction rates, mechanism of process
Week 12	14-12-2026	Adsorption & Diffusion, Fick's 1st & 2nd law
Week 13	21-12-2026	Characteristics of heterogeneous reactions

Course Evaluation Criteria:

Midterm Exam (50%) + Homework (10%) + Final Exam (40%)

Course Books:

Aydın, S., 2023. Metalurji ve Malzeme Mühendisleri için Kinetik, Literatür Yayınları.

Smith, J.M., 1981. Chemical Engineering Kinetics, 3rd Ed. Mc Graw-Hill Tokyo.

Habashi, F., 1999, Kinetics of Metallurgical Processes, Metallurgie Extractive Quebec.

Ray, S.H., 1993. Kinetics of Metallurgical Reactions, International Science Publisher.

Sohn, H.Y., Wadsworth, M.E., 1979. Rate Processes of Extractive Metallurgy, Plenum Press, New York.

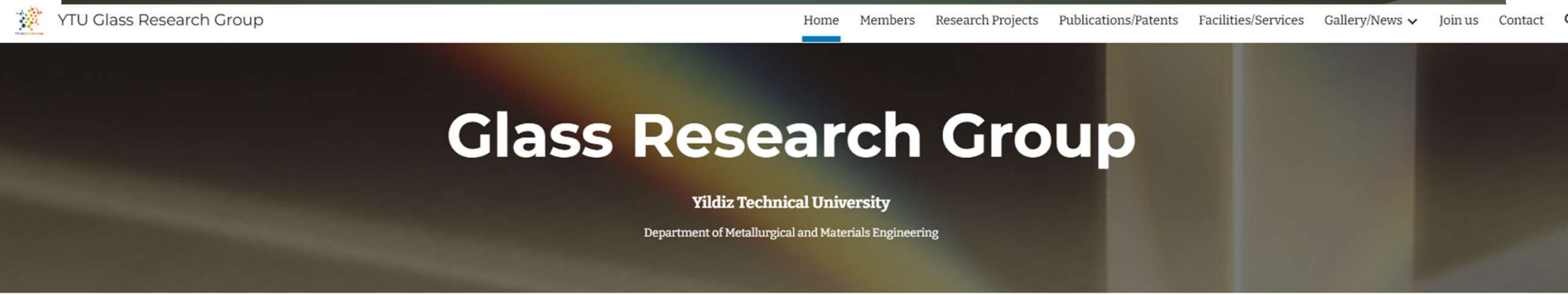
Askeland, D.R., 1984. The Science and Engineering of Materials, Brooks/Cole Engineering Division.

Geiger,G.H., Poirier,D.E., 1973. Transport Phenomena in Metallurgy, Addison-Wesley Publishing Co.

Levenspiel, O., 1972. Chemical Reaction Engineering, 2.nd Ed., John Wiley and Sons. Inc., USA.

O'Hayre, R., Materials Kinetics Fundamentals, John Wiley & Sons, Inc., Hoboken, New Jersey.

Ders Dokümanları



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Design of a metalurgical process

The most important questions while designing a metallurgical reactor and process are

1. **Which reactions will be realized in the reactor?**
2. **What will be the rate of these reactions until they reach the equilibrium?**

The first question is related with materials thermodynamics while material kinetics is interested with the second one.



Kinetic Analyses of Metallurgical Systems

In addition to the feasibility of a reaction, thermodynamics can predict the compositions of co-existing phases under certain pressure and temperature.

Thermodynamics is interested only in the initial and final states of a transformation process without considering the **reaction path, mechanism of conversion and the time required**.

What path a reaction takes, would again depend over number of details as well as properties of the system that are structure-sensitive. Therefore, **kinetic predictions are mostly either qualitative or quantitative** with some uncertainty.

Kinetic Analyses of Metallurgical Systems

Two questions about a chemical reaction are of paramount importance to engineers engaged in research, design or operation involving chemical reactions:

1. Under what conditions a chemical reaction will be feasible and how far can the reaction proceed.
2. How fast a reaction would take place under some desired operating conditions.

In general, the subject which deals with the measurement of reaction rate is turned as "Reaction Kinetics".

Metalurji Mühendisliği - Kinetik

- Proses içindeki olayları planlayabilmesi, şartları iyileştirebilmesi veya prosese müdehale edebilmesi için **reaksiyon mekanizmasının bilmesi** gereklidir.
- Metalurjide, yüksek basınç ve sıcaklık, zehirli gazlar veya korozif ortam gibi nedenlerle kapalı sistemlerde çalışılır, dolayısıyla olayların gözlemlenebilmesi sınırlıdır ya da mümkün değildir.
- Uygulama esnasında reaktan reaksiyon bitmeden reaktörü terk etmemelidir. **Reaktanların reaktörde kalış süresi, dolayısıyla reaktörün büyüklüğü kinetik deneyler sonucu elde edilen zaman bağıntıları ile planlanır.**
  - **Reaksiyon hızlı ise sadece termodinamik veriler yeterlidir.**
  - **Reaksiyon hızı düşük ise kinetik veriler de gereklidir.**

Metallurgical Engineering - Kinetic

- In order to plan the events in the process, improve the conditions or intervene in the process, it is necessary to know **the reaction mechanism**.
- In metallurgy, closed systems are used due to reasons such as high pressure and temperature, toxic gases or corrosive environments, so the observation of events is limited or impossible.
- During the application, the reactant should not leave the reactor before the reaction is completed. **The residence time of the reactants in the reactor, and therefore the size of the reactor, is planned with the time relations obtained as a result of kinetic experiments.**
  - **If the reaction is fast, only thermodynamic data is sufficient.**
  - **If the reaction rate is low, kinetic data is also necessary.**

The same crystal - Different timescales

Geological history

Diamonds formed deep within Earth under specific pressure, temperature and chemical conditions.



Hours to weeks

HPHT growth uses controlled pressure, temperature, a carbon source and a diamond seed.

What determines how quickly the crystal grows?

If pressure and temperature matter, would doubling either one always double the growth rate?

Reaction Rate

The reaction rate (rate of reaction) or speed of reaction for a reactant or product in a particular reaction is defined as how fast or slow a reaction takes place. It depends on the concentration (quantitative / extensive properties) of the reactants.

$$a\mathrm{A} + b\mathrm{B} \rightarrow c\mathrm{C}$$

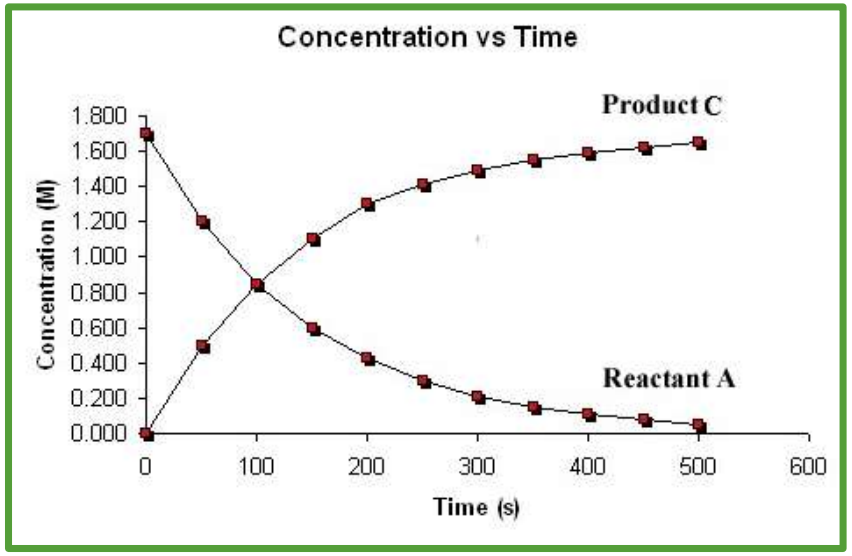
The lowercase letters (*a*, *b*, and *c*) represent stoichiometric coefficients, while the capital letters represent the reactants (A and B) and the product (C).

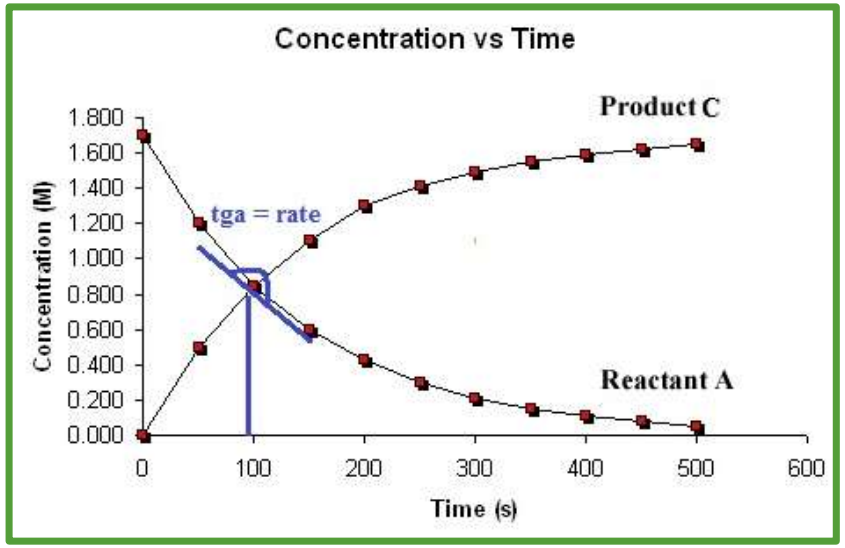
The rate of a reaction is always positive. A negative sign is present to indicate the reactant concentration is decreasing

$$\mathrm{c} = \frac{n}{V} \qquad r = -\frac{dc}{dt} \qquad x = C_0 - C \qquad r = \frac{dx}{dt}$$

Reaction Rate

***a*A + *b*B → *c*C**





t = 0	t = t
C_o	C

$$r = -\frac{dc}{dt} \qquad r = \frac{c - c_0}{t - t_0}$$

$$c_0 - c = x$$

***x*: reacted part of the reactant / product**

Reaction Rate

$aA + bB \rightarrow cC$

r = f(T, C, P)
P(constant)

$$r = -\frac{dc_A}{dt} = -\frac{dc_B}{dt} = \frac{dc_C}{dt}$$

k: reaction rate coefficient / rate constant
k = f(T)

$$r_A = -\frac{dc_A}{dt} = k c_A^a c_B^b$$

B + 2D → 3T

$$r_B = -k' c_B c_D^2 \qquad -r_B = -\frac{1}{2} r_D = \frac{1}{3} r_T$$

$$r_D = -k'' c_B c_D^2$$

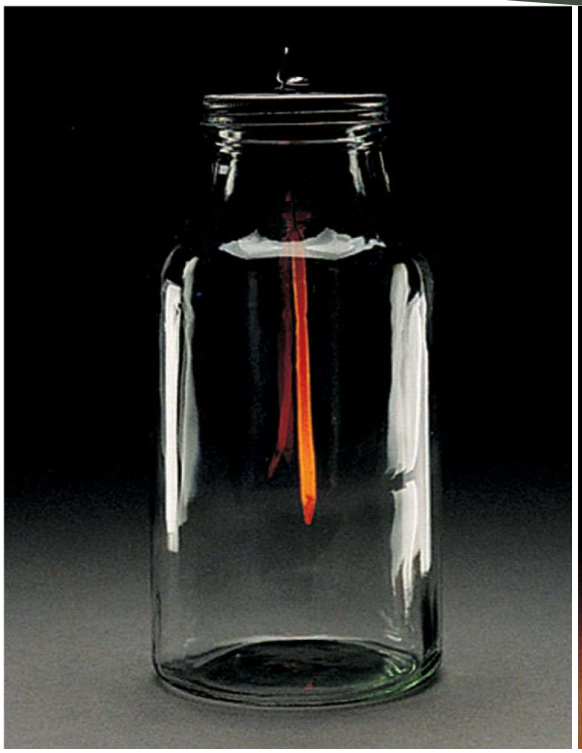
$$r_T = k''' c_B c_D^2 \qquad k' = \frac{1}{2} k'' = \frac{1}{3} k'''$$

Factors Affecting Reaction Rates

Chemical reaction rates (meaning mass of reactants transformed per unit time) generally depend on the following:

- **Concentrations of reactants** (or more accurately the thermodynamic activities)
- **Temperature of reaction** – increase in temperature increases reaction rate
- **Physical state of reactants** – whether in a gaseous phase or in a solution (homogeneous reactions) or in separate phases (heterogeneous reaction as in a sponge iron rotary kiln) – whether in solid blocks (low surface area) or in powder, colloidal or emulsion form (high surface area). Higher the surface area, higher are the reaction sites and therefore higher are the reaction rates.
- **Presence of Catalyst or Inhibitor**
- **Means of excitation** – Light, Ultraviolet or Radioactive Radiation, etc.

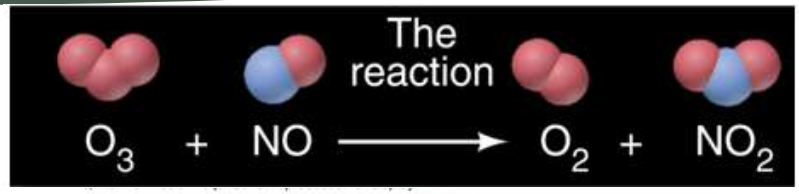
Factors Affecting Reaction Rates

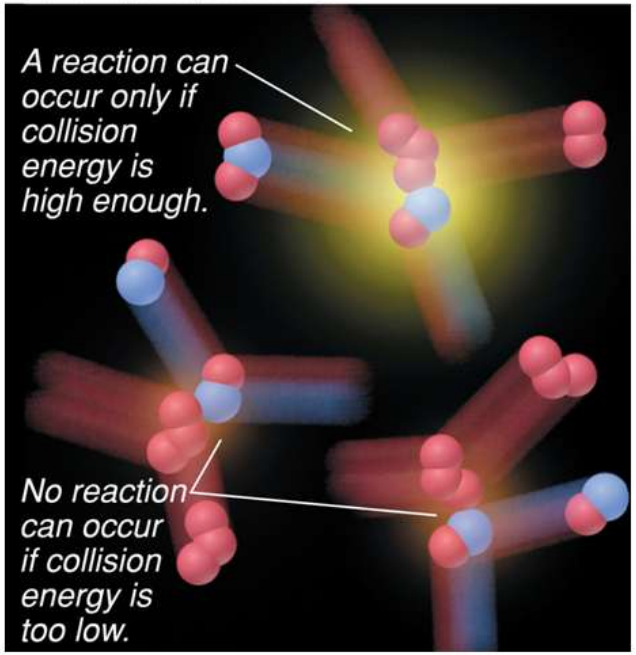


A hot steel nail glows feebly when placed in O_2.



The same mass of steel wool bursts into flame.





Classification of Reactions

There are several ways of classifying chemical reactions.

- **The basis of mechanism, with respect to the number of molecules actually participation in the reaction, with respect to the operating conditions.**
- **However, the most useful scheme is possibly according to the number and types of phases involved in a chemical reaction.**

Phase	Molecularity	Degree	Mechanism	Catalysis
Homogeneous	One molecular	Zeroth degree	Reversible	Catalytic
Heterogeneous	Two molecular	First degree	Irreversible	Non-catalytic
	Multi molecular	Second degree	Parallel	
		nth degree	Serial	

Classification of Reactions

A reaction is **homogeneous** if it occurs in a single phase. Most gas phase reactions are examples of homogeneous reactions. If a reaction requires the presence of two phases, it is **heterogeneous**. The burning of coal, the roasting of ores, and the dissolution of solids in acid are all heterogeneous reactions.

It is to be rated that these classifications are not mutually exclusive. Thus a reaction may be irreversible-second order, Taking place under adiabatic and approximately constant pressure conditions in a continuous type reactor.

They may be **irreversible, reversible, simultaneous or consecutive** type with respect to the number of molecules actually participating in the reaction, it may **unimolecular or bimolecular**. Considering the operating conditions, a chemical reaction maybe **isotherma**l at constant pressure, **adiabatic or non-adiabatic** and **non-isotherma**l.

Reaksiyonların Sınıflandırılması

Fazlarına göre	Molekülaritesine göre	Derecesine göre	Mekanizmasına göre	Katalizli oluşuna göre
Homojen	Bir moleküllü	Sıfırıncı derece	Dönüşümsüz	Katalitik
Heterojen	İki moleküllü	Birinci derece	Dönüşümlü (tersinir)	Katalitik olmayan
	Çok moleküllü	İkinci derece	Paralel	
		n. derece	Seri	

According to the phases

Homogeneous Reactions: it takes place in one phase along (reactants and products)

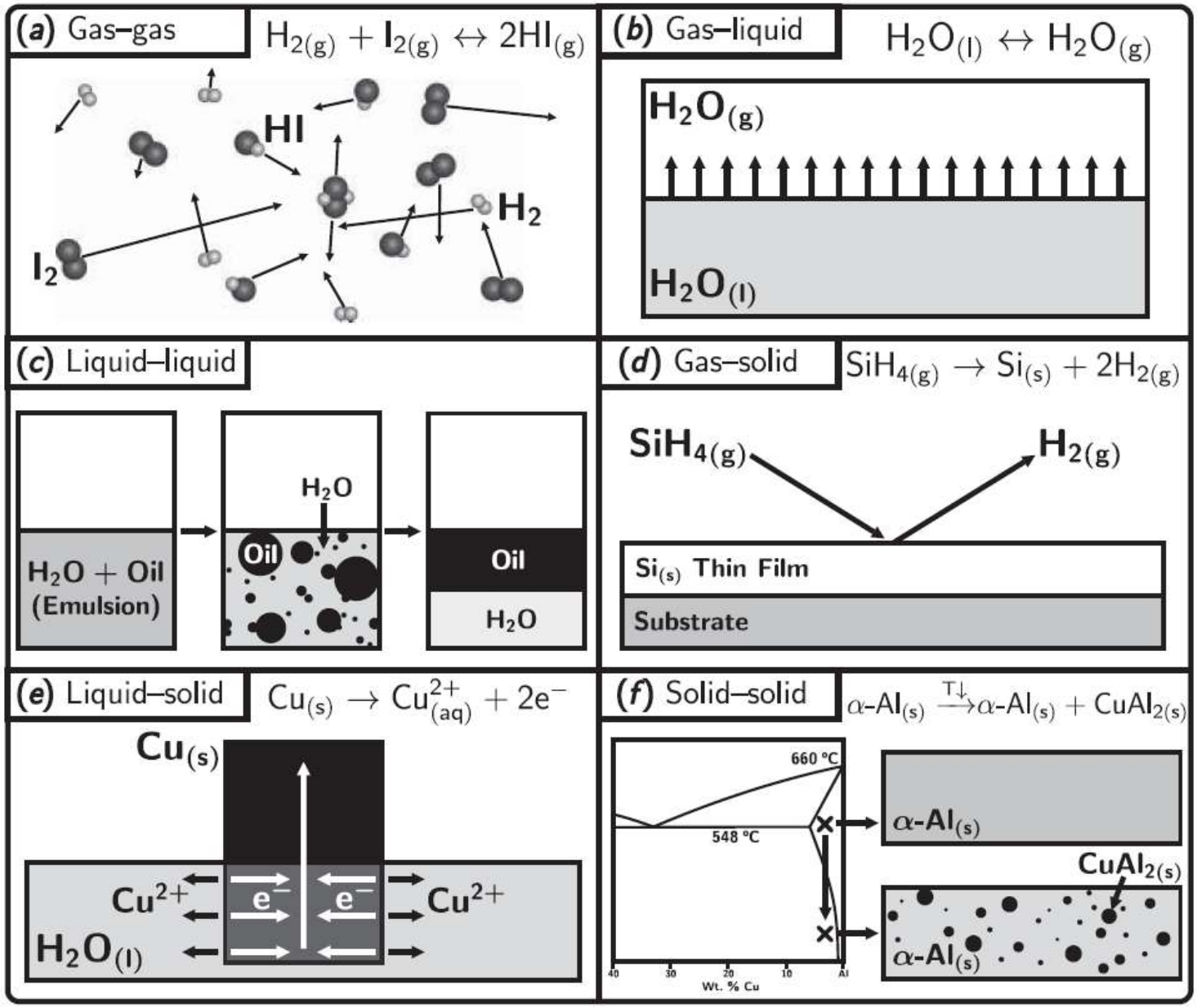
$$CO_{(g)} + \tfrac{1}{2} O_{2(g)} = CO_{2(g)}$$

Heterogeneous Reactions: minimum two different phase are required

$$C_{(s)} + O_{2(g)} = CO_{2(g)}$$

All pyrometallurgic reactions, solid-gas, gas-liquid-metal, gas-oxide-liquid reactions are heterogeneous.

S-S, S-L, S-G, L-L, L-G





a) Gas–gas: reaction equilibrium between hydrogen gas, iodine gas, and hydrogen iodide gas.

b) Gas–liquid: evaporation of liquid water from a glass.

c) Liquid–Liquid: gradual separation of an oil–water mixture.

d) Gas–solid: chemical vapor deposition of a thin Si film.

e) Liquid–solid: corrosion of Cu metal in seawater.

f) Solid–solid: precipitation of $CuAl_2$ particles from a copper–aluminum alloy during a heat treatment process.

According to the order

A rate law is a differential equation that describes **the rate of change of a reactant (or product) concentration with time.**

If we integrate the rate law then we obtain an expression for the concentration as a function of time, which is generally the type of data obtained in an experiment.

Reaksiyon derecesi

Reaktanların (ya da ürünlerin) konsantrasyonun değişiminin zamana kaçıncı dereceden bağlı olduğudur.

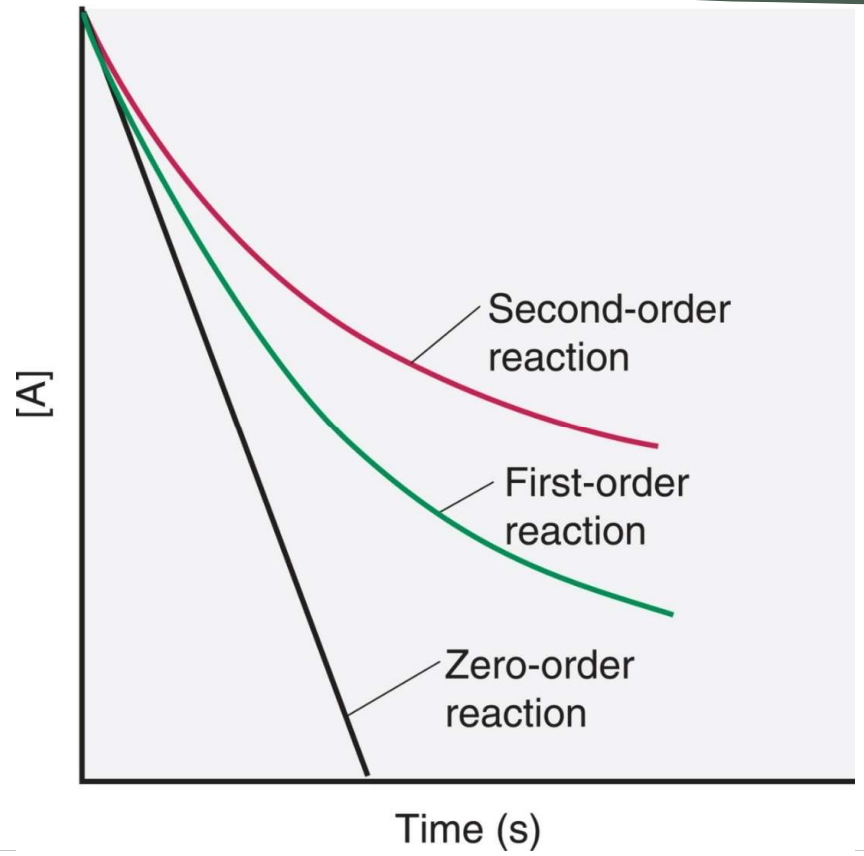
Order of Reaction

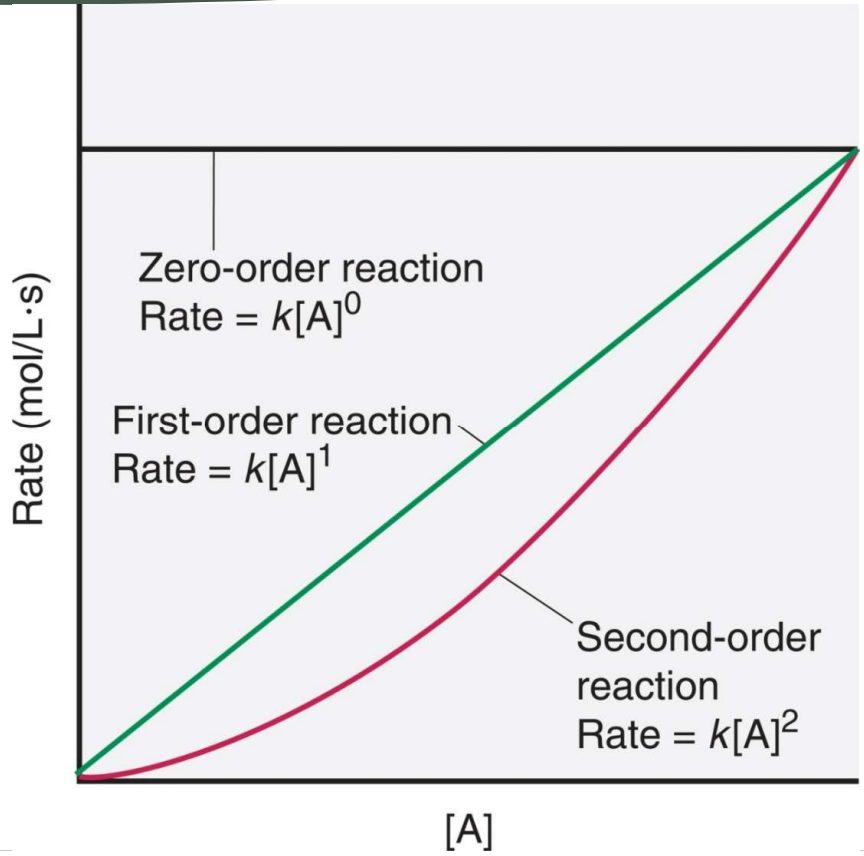
The **order of reaction** with respect to a given substance (such as reactant, catalyst or product) is defined as the index, or exponent, to which its concentration term in the rate equation is raised.

A rate law is a differential equation that describes **the rate of change of a reactant (or product) concentration with time**. If we integrate the rate law then we obtain an expression for the concentration as a function of time, which is generally the type of data obtained in an experiment. In many simple cases, the rate law may be integrated analytically.

- For **zero-order reactions**, the reaction rate is independent of the concentration of a reactant, so that changing its concentration has no effect on the speed of the reaction.
- If a reaction rate depends on a single reactant and the value of the exponent is one, then the reaction is said to be **first order**.
- A reaction is said to be **second order** when the overall order is two. The rate of a second-order reaction may be proportional to one concentration squared or (more commonly) to the product of two concentrations

Order of Reaction





Homogeneous Reaction Kinetics

αA + βB + γC = ...

$$r = kc_A^{\alpha} c_B^{\beta} c_C^{\gamma}$$

	t = 0	t = t
C_A	a	x
C_B	b	y
C_C	c	z

reaction rate at time t

$$r = k(a\text{-}x)^{\alpha}(b\text{-}y)^{\beta}(c\text{-}z)^{\gamma}$$

If a = b = c x = y = z

$$r = k(a\text{-}x)^{\alpha+\beta+\gamma}$$

$$r = k(a\text{-}x)^{n}$$

$$\mathbf{r} = \frac{\boldsymbol{dx}}{\boldsymbol{dt}} = \boldsymbol{k}(\boldsymbol{a} - \boldsymbol{x})^{\boldsymbol{n}}$$

reaction order

reaction rate

rate constant

Order of Reaction

$Ra \rightarrow Rn + He$	$r = k[Ra]^1 = k[Ra]$	1st order
$2HI \rightarrow H_2 + I_2$	$r = k[HI]^2$	2nd order
$2NO + O_2 \rightarrow 2NO_2$	$r = k[NO]^2[O_2]$	3rd order

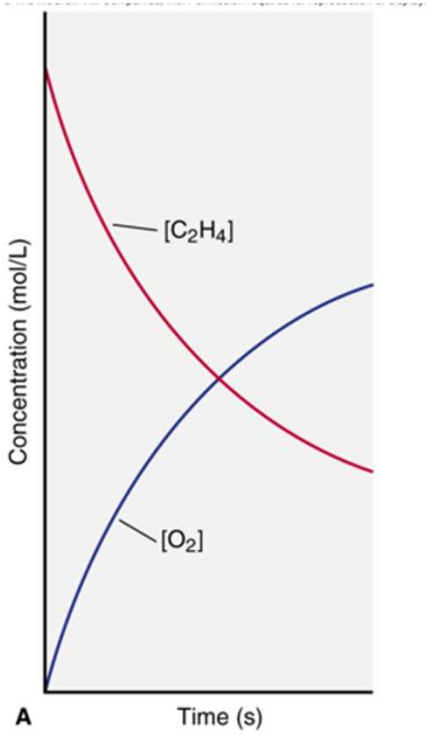
Three types of reaction rates for the reaction of O_3 and C_2H_4

$$C_2H_4\,(g) + O_3\,(g) \rightleftharpoons C_2H_4O\,(g) + O_2\,(g)$$

Time (s)	Concentration of O_3 (mol/L)
0.0	3.20×10^{-5}
10.0	2.42×10^{-5}
20.0	1.95×10^{-5}
30.0	1.63×10^{-5}
40.0	1.40×10^{-5}
50.0	1.23×10^{-5}
60.0	1.10×10^{-5}

$$\text{rate} = -\frac{\Delta [C_2H_4]}{\Delta t}$$

$$= -\frac{\Delta [O_3]}{\Delta t}$$



$[O_2]$ increases just as fast as $[C_2H_4]$ decreases.

$$\text{Rate} = -\frac{\Delta[C_2H_4]}{\Delta t} = -\frac{\Delta[O_3]}{\Delta t}$$

$$= \frac{\Delta[C_2H_4O]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t}$$

Zeroth Order Reactions

Reaction rate is independent of reactant concentration

$$\mathbf{r} = \frac{dx}{dt} = k_0(a-x)^0 = k_0$$

$$\int dx = \int k_0 dt$$

$$x_0 = k_0 t + cst$$

If t = 0, x = 0 cst = 0

$$\boxed{x_0 = k_0 t}$$

k_0 = M.sec^{-1}

Zeroth order half life

$$x = k_0 t$$

$$x = \frac{a}{2}$$

$$\frac{a}{2} = k_0 t_{1/2} \qquad \boxed{t_{1/2} = \frac{a}{2k_0}}$$

Half-life is proportional with the concentration

Zeroth Order Reactions

In a zero-order reaction, increasing the concentration of the reactant(s) will not speed up the rate of the reaction.

The most common zero-order reactions are endothermic high-temperature thermal decomposition reactions, where a large amount of thermal energy is required to break a chemical species apart.

In such cases, as long as the temperature is not too high, the available thermal energy controls the rate rather than the concentration of the reactant.

Sıfırıncı dereceden bir reaksiyonda, reaktanların konsantrasyonunun arttırılması, reaksiyonun hızını artırmayacaktır. Sıcaklık çok yüksek olmadığı sürece, reaktanların konsantrasyonu değil mevcut termal enerji hızı kontrol eder.

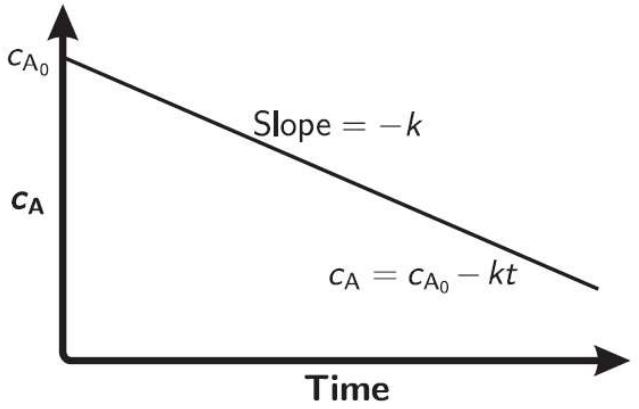


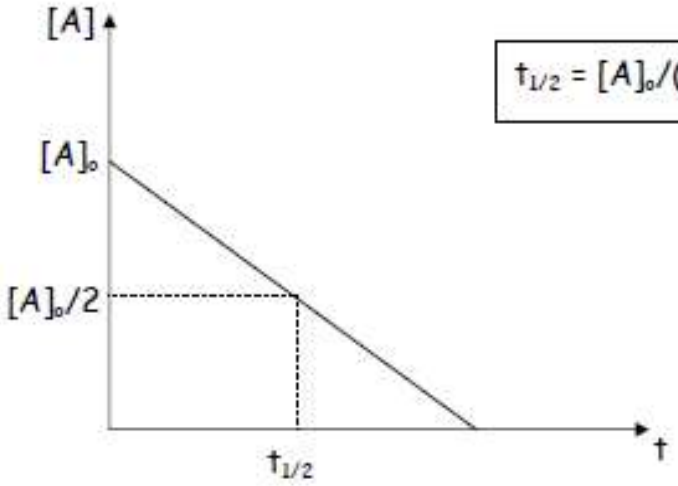
FIGURE 3.2 Variation in reactant concentration as a function of time for a zero-order reaction that consumes species A. The concentration of species A decreases linearly with time from an initial concentration c_{A_0}. The slope is given by the rate constant k.

I) Zero Order Reactions (rare)

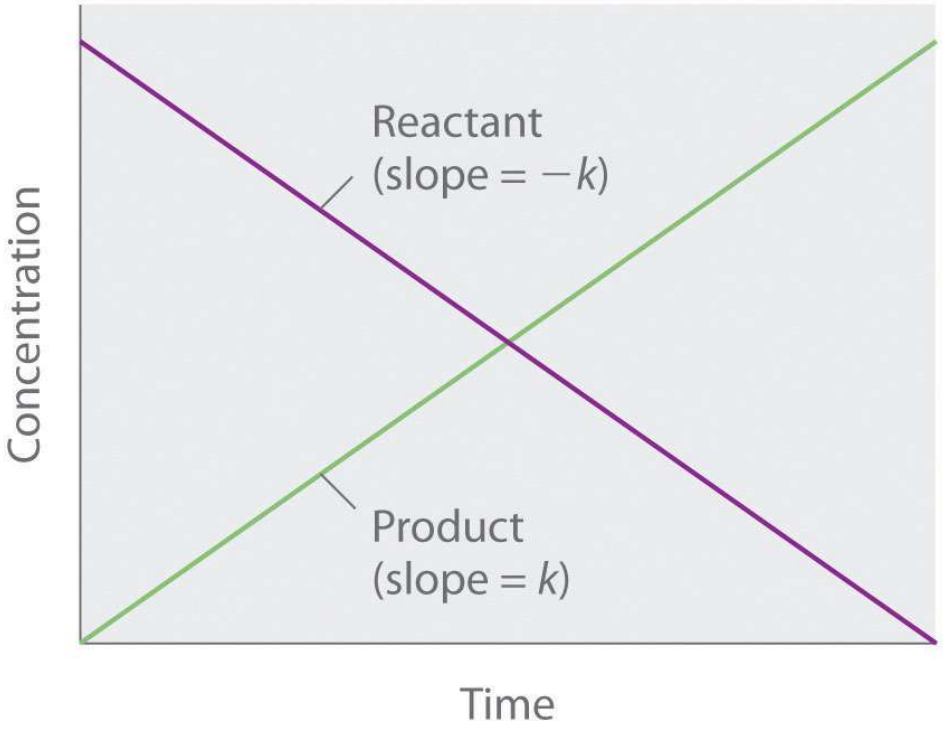
$A \rightarrow$ products

$$-\frac{d[A]}{dt} = k \qquad \{k \text{ is in [moles/(liter sec)]}\}$$

$$[A] = -kt + [A]_o$$



$$t_{1/2} = [A]_o/(2k)$$



In Zero Order Reactions,

- **Reaction rate is independent of concentration**
- **Reaction rate vs. time show a lineer dependence**
- **Reaction rate is independent of time**

An example is the decomposition of N_2O on a platinum (Pt) surface to produce N_2 and O_2, which occurs at temperatures ranging from 200°C to 400°C:

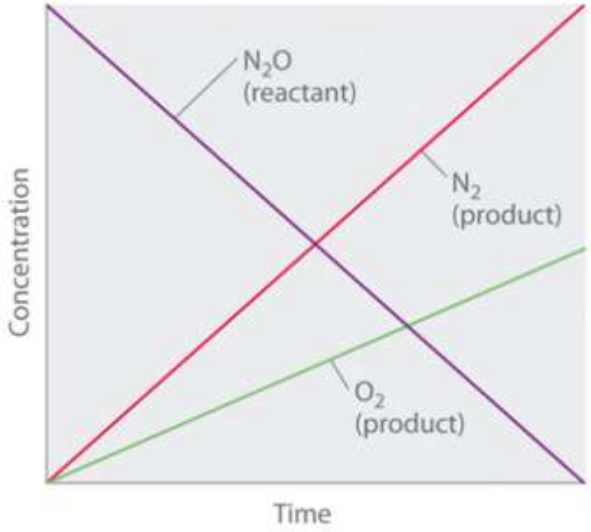
$$2N_2O(g) \xrightarrow{Pt} 2N_2(g) + O_2(g)$$

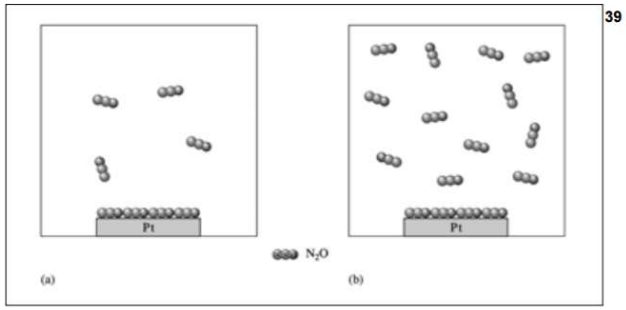
Without a platinum surface, the reaction requires temperatures greater than 700°C, but between 200°C and 400°C, the only factor that determines how rapidly N_2O decomposes is the amount of Pt surface available (not the amount of Pt). As long as there is enough N_2O to react with the entire Pt surface, doubling or quadrupling the N_2O concentration will have no effect on the reaction rate. At very low concentrations of N_2O, where there are not enough molecules present to occupy the entire available Pt surface, the reaction rate is dependent on the N_2O concentration. The reaction rate is as follows:

$$\text{rate} = -\frac{1}{2}\left(\frac{\Delta[N_2O]}{\Delta t}\right) = \frac{1}{2}\left(\frac{\Delta[N_2]}{\Delta t}\right) = \frac{\Delta[O_2]}{\Delta t} = k[N_2O]^0 = k$$

Thus the rate at which N_2O is consumed and the rates at which N_2 and O_2 are produced are independent of concentration. As shown in Figure 14.8, the change in the concentrations of all species with time is linear. Most important, the exponent (0) corresponding to the N_2O concentration in the experimentally derived rate law is not the same as the reactant's stoichiometric coefficient in the balanced chemical equation (2). For this reaction, as for all others, the rate law must be determined experimentally.

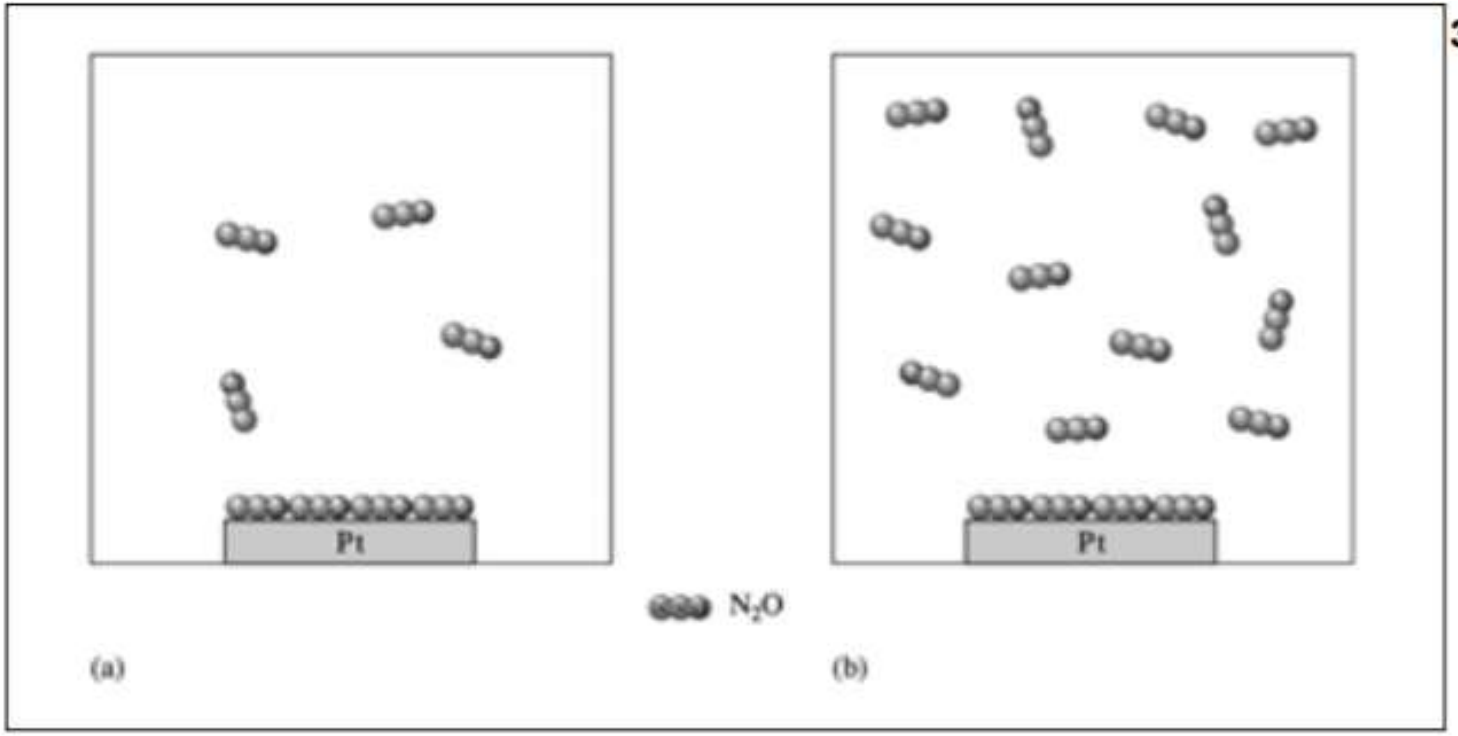
Zeroth order reactions occur most often when the reaction rate is determined by available surface area





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- **The platinum surface is completely covered with N_2O molecules.**
- **If we increase the concentration, it has no effect on the rate.**
- **Since only those N_2O molecules on the surface can react.**
- **An Example of a zero-order reaction.**



- The platinum surface is completely covered with N_2O molecules.
- If we increase the concentration, it has no effect on the rate.
- Since only those N_2O molecules on the surface can react.
- An Example of a zero-order reaction.

Zeroth Order Reactions: Reverse Haber Process

$$2NH_{3(g)} \rightarrow N_{2(g)} + 3H_{2(g)}$$

$$\frac{dc_{NH_3}}{dt} = -k(c_{NH_3})^0 = -k$$

c_A is the concentration of species A at time t and c_{A0} is the initial concentration of species A at time $t = 0$.

$$\frac{dc_A}{dt} = -k$$

$$\int dc_A = -k \int dt$$

$$c_A - c_{A_0} = -kt$$

$$c_A = c_{A_0} - kt$$

First Order Reactions

Reaction rate is proportional to the amount of converted material

$$\mathbf{r} = \frac{dx}{dt} = k_1(a-x)^1$$

If a - x = z, -dx = dz

$$\frac{dx}{a-x} = k_1 dt \qquad \frac{-dz}{z} = k_1 dt$$

$$-lnz = k_1 t + cst$$

$$-ln(a-x) = k_1 t + cst \quad \text{t=0, x=0} \quad \text{cst = -lna}$$

$$-ln(a-x) = k_1 t - lna$$

$$ln\frac{a}{a-x} = k_1 t$$

$$k_1 = \frac{1}{t} ln\frac{a}{a-x}$$

$$k_1 = \frac{2.303}{t} log\frac{a}{a-x}$$

$$k_1 = \frac{2.303}{t} log\frac{C_0}{C}$$

$k_1 = sec^{-1}$

First Order Reactions

$$k_1 = \frac{1}{t} ln \frac{a}{a - x}$$

First order half life

$$x = \frac{a}{2}$$

$$t_{1/2} = \frac{1}{k_1} ln \frac{a}{a - \frac{a}{2}} = \frac{1}{k_1} ln2$$

$$\boxed{t_{1/2} = \frac{0.693}{k_1}}$$

Half-life is independent of the first concentration

Even if the initial concentration differs, the time required for the amount of substance to decrease by half is the same!!!

First Order Reactions

$$\frac{dc_A}{dt} = -kc_A$$

$$\int \frac{dc_A}{c_A} = -k \int dt$$

$$\ln \frac{c_A}{c_{A_0}} = -kt$$

$$c_A = c_{A_0} e^{-kt}$$

In a first-order reaction, the reaction rate is directly proportional to the concentration of one of the reactant concentrations. Thus, increasing the concentration of this reactant will speed up the rate of the reaction proportionally.

This behavior reflects the fundamental kinetic principle that the speed of most reactions depends on the frequency with which the reactants encounter one another.

Birinci dereceden bir reaksiyonda, reaksiyon hızı, reaktanlardan birinin konsantrasyonu ile doğru orantılıdır.

Bu reaktanın konsantrasyonunun arttırılması, reaksiyon oranını orantılı olarak hızlandıracaktır.

$$\frac{dc_A}{dt} = -kc_A^1 = -kc_A$$

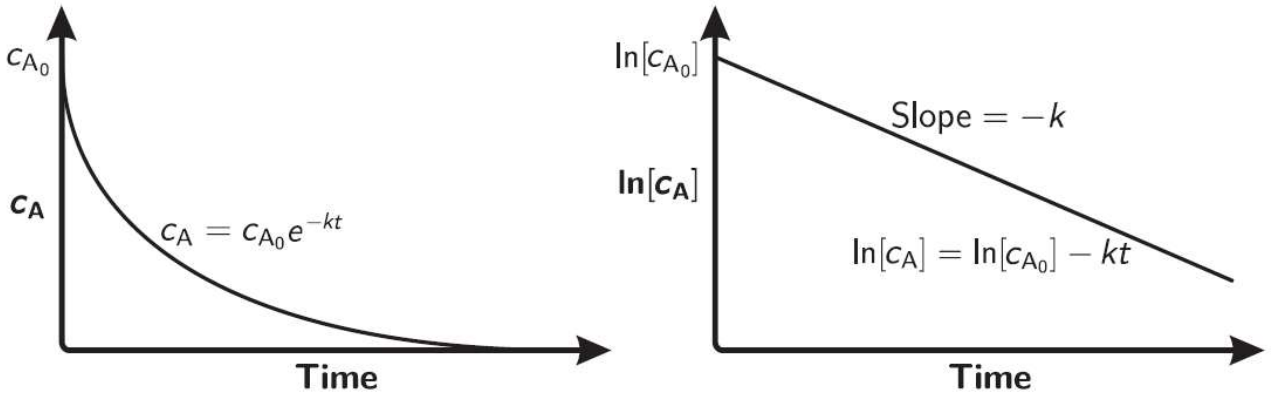


FIGURE 3.3 Variation in reactant concentration as a function of time for a first-order reaction that consumes species A. (*a*) The concentration of species A decreases exponentially with time from an initial concentration c_{A_0}. (*b*) A logarithmic plot of concentration versus time yields a straight line with a slope that is given by the rate constant k.

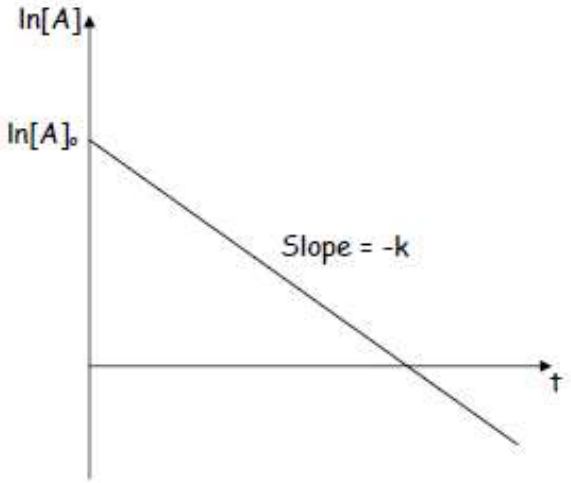
II) First Order Reactions

$A \rightarrow$ products

$$-\frac{d[A]}{dt} = k[A]$$ {k is in [1/sec]}

$$[A] = [A]_o e^{-kt}$$

$$\ln[A] = -kt + \ln[A]_o$$



$$t_{1/2} = (\ln 2) / k = 0.693/k$$

$$k = (0.693)/(t_{1/2})$$

$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$

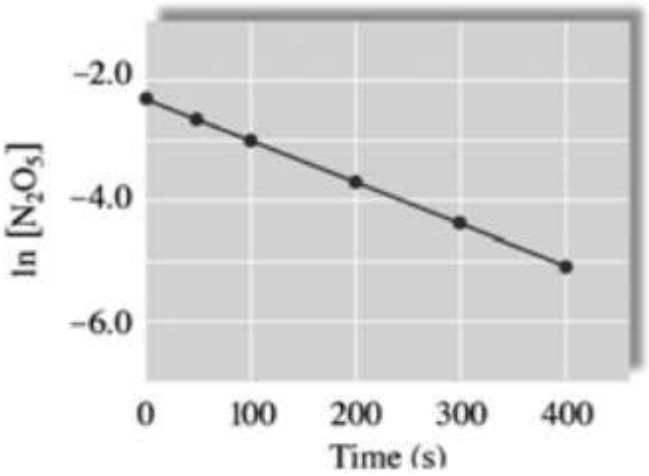
$[N_2O_5]$ (M)	Time (sec)
0.1000	0
0.0707	50
0.0500	100
0.0250	200
0.0125	300
0.00625	400

1. Verify that the rate law is first order.
2. Calculate the rate constant.

ln $[N_2O_5]$	Time (sec)
-2.303	0
-2.649	50
-2.996	100
-3.689	200
-4.382	300
-5.075	400

The plot proves it is a straight line and so it is a first order reaction.

$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$



$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$

Since it is a first order reaction, we know that in the first order integrated rate law that k is equal to the -slope of the line.

$$\text{Slope} = \frac{\Delta Y}{\Delta X} = \frac{\Delta(\ln[N_2O_5])}{\Delta t}$$

$$\text{Slope} = \frac{-5.075 - (-2.303)}{400 - 0} = \frac{-2.772}{400} = -6.93 \times 10^{-3}\ s^{-1}$$

$$k = -(\text{slope}) = 6.93 \times 10^{-3}\ s^{-1}$$

First Order Reactions

The half-life for the radioactive decay of carbon-14 is 5730 years. Calculate the rate constant for this decay reaction.

$$t_{1/2} = \frac{0.693}{k}$$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{5730\ \text{yr}}$$

$$k = 1.21 \times 10^{-4}/\text{yr} = 2.30 \times 10^{-10}/\text{min} = 3.83 \times 10^{-12}/\text{s}$$

Second Order Reactions

Reaction rate is proportional with the concentration squared

$$\mathrm{r} = \frac{dx}{dt} = k_2(a-x)^2$$

$$\frac{dx}{(a-x)^2} = k_2 dt$$ If a - x = z, -dx = dz

$$\frac{-dz}{z^2} = k_2 dt$$

$$\int \frac{-dz}{z^2} = \int k_2 dt$$

$$\frac{1}{z} = k_2 t + cst$$

$$\frac{1}{a-x} = k_2 t + cst$$ t = 0, x = 0, cst = $\frac{1}{a}$

$$\frac{1}{a-x} = k_2 t + \frac{1}{a}$$

$$k_2 = \frac{1}{t}\frac{x}{a(a-x)}$$

$$k_2 = \frac{1}{t}\frac{C_0 - C}{C_0 * C}$$

$k_2 = M^{-1}.sec^{-1}$

Second Order Reactions

Second order half life

$$x = \frac{a}{2}$$

$$t = \frac{1}{k_2} \frac{x}{a(a-x)}$$

$$t_{1/2} = \frac{1}{k_2} \frac{\frac{a}{2}}{a(a-\frac{a}{2})}$$

$$\boxed{t_{1/2} = \frac{1}{k_2 a}}$$

Half-life is inversely proportional to the concentration

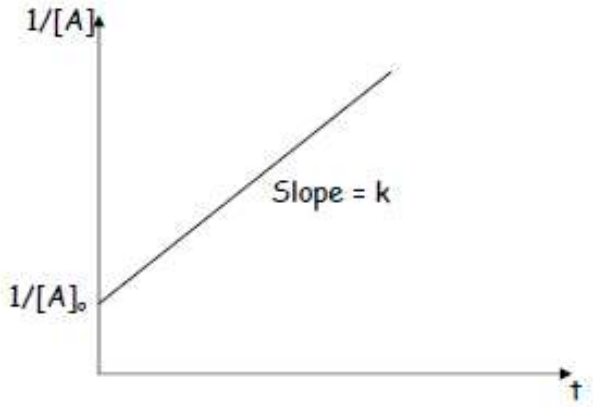
III) Second Order Reactions

a) Second order in one component

$A \rightarrow$ products

$$-\frac{d[A]}{dt} = k[A]^2$$

$$\boxed{\frac{1}{[A]} = \frac{1}{[A]_o} + kt}$$



$$\boxed{t_{1/2} = 1/(k[A]_o)}$$

Second Order Reactions

$$2NO_2(g) \rightarrow 2NO(g) + O_2(g)$$

The decomposition of NO_2 to NO and O_2, has been experimentally determined to be second order with respect to the NO_2 concentration.

A reaction begins at time t = 0 with an unknown initial concentration of pure $NO_2(g)$.

After 2 h, the rate of O_2 gas production is measured as 1.0×10^{-10} mol/(cm^3·s).

If the reaction rate constant for this reaction is k = 10 (mol/cm^3 · s)$^{-1}$, determine what the initial concentration of $NO_2(g)$ must have been when the reaction began.

$$\frac{dc_{NO_2}}{dt} = -k(c_{NO_2})^2$$

$$\frac{1}{c_{NO_2}} = \frac{1}{c_{NO_2O}} + kt$$

$$\left.\frac{dc_{NO_2}}{dt}\right|_{t=2\,h} = -2\left.\frac{dc_{O_2}}{dt}\right|_{t=2\,h}$$

$$\left.\frac{dc_{NO_2}}{dt}\right|_{t=2\,h} = -2\left.\frac{dc_{O_2}}{dt}\right|_{t=2\,h} = -k(c_{NO_2}|_{t=2\,h})^2$$

$$c_{NO_2}|_{t=2\,h} = \sqrt{\left(\frac{2}{k}\right)\left.\frac{dc_{NO_2}}{dt}\right|_{t=2\,h}}$$

$$= \sqrt{\left(\frac{2}{10\ (\text{mol/cm}^3 \cdot \text{s})^{-1}}\right) 1.00 \times 10^{-10}\ \text{mol/(cm}^3 \cdot \text{s)}}$$

$$= 4.47 \times 10^{-6}\ \text{mol/cm}^3$$

$$\frac{1}{c_{NO_2}|_{t=2\,h}} = \frac{1}{c_{NO_{20}}|_{t=2\,h}} + kt$$

$$c_{NO_{20}} = \frac{c_{NO_2}|_{t=2\,h}}{1 - c_{NO_2}|_{t=2\,h} kt}$$

$$= \frac{4.47 \times 10^{-6}\ \text{mol/cm}^3}{1 - 4.47 \times 10^{-6}\ \text{mol/cm}^3 \cdot 10\ (\text{mol/cm}^3 \cdot \text{s})^{-1} \cdot 7200\ \text{s}}$$

$$= 6.59 \times 10^{-6}\ \text{mol/cm}^3$$

Third and nth Order Reactions

Third order

$$\mathrm{r} = \frac{dx}{dt} = k_3(a-x)^3$$

$$k_3 = \frac{x(2a-x)}{2ta^2(a-x)^2}$$

$$t_{1/2} = \frac{3}{2k_3a^2}$$

nth order

$$\mathrm{r} = \frac{dx}{dt} = k_n(a-x)^n$$

$$k_n = \frac{1}{(n-1)t}\left[\frac{1}{(a-x)^{n-1}} - \frac{1}{a^{n-1}}\right]$$

$$t_{1/2} = \frac{(2^{n-1}-1)}{(n-1)k_n a^{(n-1)}}$$

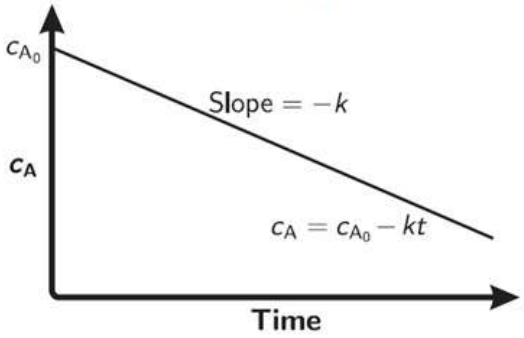
Order of Reaction

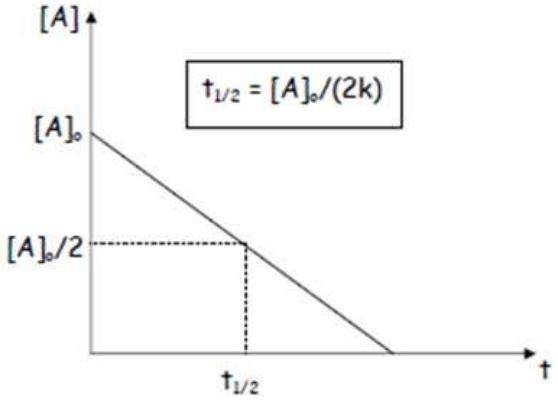
$$FeO + CO \rightarrow Fe + CO_2$$

Condition	Rate-controlling step	Observed kinetic order	Explanation
Surface saturated with CO (high CO pressure)	Surface saturation rather than surface reaction	**Zero order**	The rate is no longer dependent on [CO]; it remains constant.
Moderate CO pressure	Surface reaction–controlled	**First order**	The rate is directly proportional to [CO].
Low CO pressure, reverse reaction significant	Both forward and reverse reactions are influential	**Approximately second order (or mixed)**	The rate depends on both [CO] and [CO_2] concentrations.

Order of Reaction

zeroth-order reaction

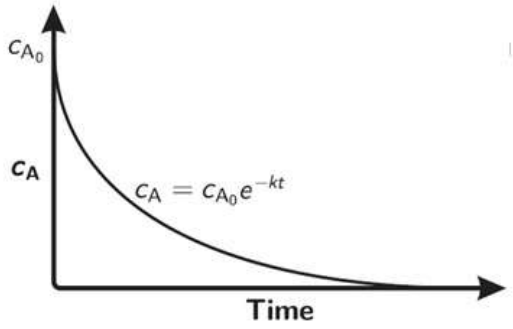


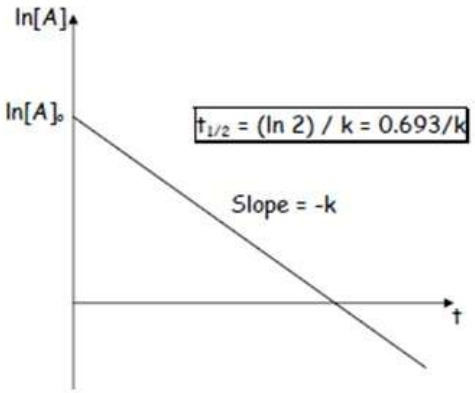


$$\text{rate} = -\frac{\Delta[A]}{\Delta t} = k$$

$[A] = [A]_0 - kt$

first-order reaction



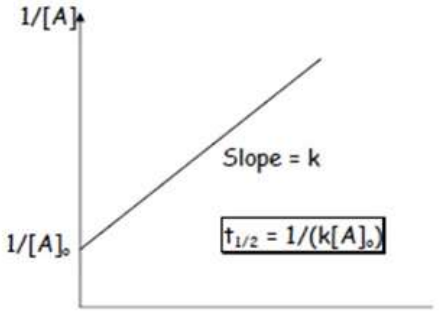


$$\text{rate} = -\frac{\Delta[A]}{\Delta t} = k[A]$$

$[A] = [A]_0 e^{-kt}$

$\ln[A] = \ln[A]_0 - kt$

second-order reaction



$$\text{rate} = -\frac{\Delta[A]}{\Delta t} = k[A]^2$$

$$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$$

Summary of the Kinetics for Reactions of the Type aA → Products that are Zero, First, or Second Order in [A]			
	Order		
	Zero	*First*	*Second*
Rate law:	Rate = k	Rate = k[A]	Rate = k[A]^2
Integrated rate law:	[A] = $-kt$ + [A]$_0$	ln[A] = $-kt$ + ln[A]$_0$	$\frac{1}{[A]} = kt + \frac{1}{[A]_0}$
Plot needed to give a straight line:	[A] versus t	ln[A] versus t	$\frac{1}{[A]}$ versus t
Relationship of rate constant to the slope of straight line:	Slope = $-k$	Slope = $-k$	Slope = k
Half-life:	$t_{1/2} = \frac{[A]_0}{2k}$	$t_{1/2} = \frac{\ln 2}{k}$	$t_{1/2} = \frac{1}{k[A]_0}$

Key Equations

zeroth-order reaction

$$\text{rate} = -\frac{\Delta[A]}{\Delta t} = k$$

$$[A] = [A]_0 - kt$$

first-order reaction

$$\text{rate} = -\frac{\Delta[A]}{\Delta t} = k[A]$$

$$[A] = [A]_0 e^{-kt}$$

$$\ln[A] = \ln[A]_0 - kt$$

second-order reaction

$$\text{rate} = -\frac{\Delta[A]}{\Delta t} = k[A]^2$$

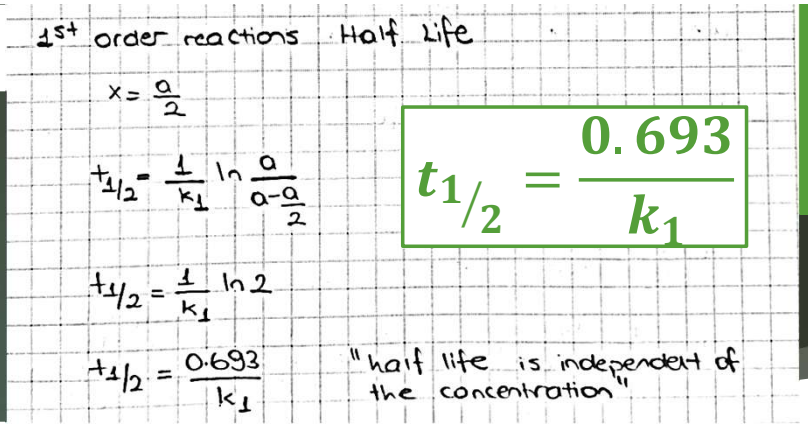
$$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$$

where $[A]_0$ is the initial concentration of reactant A at $t = 0$

TABLE 3.1 Summary of Zero-, First-, and Second-Order Reactions

			Second Order	
	Zero Order	First Order	c_A^2	$c_A c_B$
Rate law	$\frac{dc_A}{dt} = -k$	$\frac{dc_A}{dt} = -kc_A$	$\frac{dc_A}{dt} = -kc_A^2$	$\frac{dc_A}{dt} = -kc_A c_B$
Integrated	$c_A = c_{A_0} - kt$	$c_A = c_{A_0} e^{-kt}$	$\frac{1}{c_A} = \frac{1}{c_{A_0}} + kt$	$\frac{c_A}{c_B} = \frac{c_{A_0}}{c_{B_0}} \exp\left[kt\left(\frac{b}{a}c_{A_0} - c_{B_0}\right)\right]$
Units for k	$\frac{\text{concentration}}{\text{time}}$	$\frac{1}{\text{time}}$	$\frac{1}{\text{concentration} \cdot \text{time}}$	$\frac{1}{\text{concentration} \cdot \text{time}}$
Half-life	$\frac{c_{A_0}}{2k}$	$\frac{\ln 2}{k}$	$\frac{1}{kc_{A_0}}$	$-\frac{\ln\left(2 - \frac{b}{a}\frac{c_{A_0}}{c_{B_0}}\right)}{k\left(\frac{b}{a}c_{A_0} - c_{B_0}\right)}$

Example



$$t_{1/2} = \frac{0.693}{k_1}$$

Degradation of uranium is a first order reaction and its half life is $t_{1/2} = 4.51x10^9$ years. Calculate the rate constant (k) and the time required (daily) for the 75% degration of uranium.

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{4.51x10^9}$$

$$k = 1.532x10^{-1} \ \ 1/year$$

$$t = \frac{1}{k}\ln\frac{a}{a-x}$$

$$t_{0.75} = \frac{1}{k}\ln\frac{a}{a-0.75a}$$

$$= \frac{2.303x365}{1.532x10^{-1}} log4 = 330.4x10^{10} days$$

Example

$$k_2 = \frac{1}{t}\frac{x}{a(a-x)}$$

$$t_{1/2} = \frac{1}{k_2 a}$$

In a second order reaction, initial concentration of the ractants is 0.4 mol/l. At the 80th minute, 30% of reaction is realized. Calculate the rate constant and the time required (in seconds) for the 80% of reaction realized.

$$k = \frac{1}{t}\frac{x}{a(a-x)} \qquad k = \frac{1}{80x60}\frac{0.4x0.3}{0.4(0.4-0.4x0.3)} = 2.232x10^{-4}\ liter.second/mol$$

$$2.232x10^{-4} = \frac{1}{t_{0.8}}\frac{0.4x0.8}{0.4(0.4-0.4x0.8)}$$

$$t_{0.8} = \frac{0.4x0.8}{0.4(0.4-0.32)x2.232x10^{-4}} = 4.48x10^{4} seconds$$

Methods for Determining the Reaction Orders

- Integration Method
- Linearized Coordinate / Graphic Method
- Half-Life Method
- Van't Hoff Differential Method

Integration Method – Reaction Rate

Differential equation that describes **the rate of change of a reactant (or product) concentration with time**. **If we integrate the rate law then we obtain an expression for the concentration as a function of time, which is generally the type of data obtained in an experiment**.

Reaction	Order	Differential form	Integrated form
$A \rightarrow P$	zeroth	$\frac{d[A]}{dt} = -k$	$[A] = [A]_0 - kt$
$A \rightarrow P$	first	$\frac{d[A]}{dt} = -k\,[A]$	$\ln[A] = \ln[A]_0 - kt$
$A + A \rightarrow P$	second	$\frac{1}{2}\frac{d[A]}{dt} = -k\,[A]^2$	$\frac{1}{[A]} = \frac{1}{[A]_0} + 2kt$
$A + B \rightarrow P$	second	$\frac{d[A]}{dt} = -k\,[A][B]$	$kt = \frac{1}{[B]_0 - [A]_0} \ln\frac{[B]_0[A]}{[A]_0[B]}$

In the above $[A]_0$ and $[B]_0$ represent the initial concentrations of A and B i.e. their concentrations at the start of the reaction.

If I plot the concentration, [A], as a function of time for each expression below, the correct order should yield a straight line function.

Zero order	First order	Second order
$y = m\,x + b$	$y = m\,x + b$	$y = m\,x + b$
$[C] = -ck\,t + [C_0]$	$\ln[C] = -ck\,t + \ln[C_0]$	$1/[C] = ck\,t + 1/[C_0]$

İntegrasyon Methodu

Deneysel olarak saptanan konsantrasyon ve zaman değerleri daha önce farklı reaksiyon dereceleri için saptanan eşitliklerde yerine koyularak, bu değerlerin (x ve t) hangi dereceden reaksiyona uygun olduğu araştırılarak reaksiyon derecesi belirlenir.

Reaksiyon derecesi: n
Başlangıç konsantrasyonu: C_0
t sürede reaksiyona giren kısım: x

$$-\frac{d(C_0 - x)}{dt} = \frac{dx}{dt} = k(C_0 - x)^n$$

Tablo I- Farklı Reaksiyon Dereceleri için Hız Eşitliklerinin Özeti

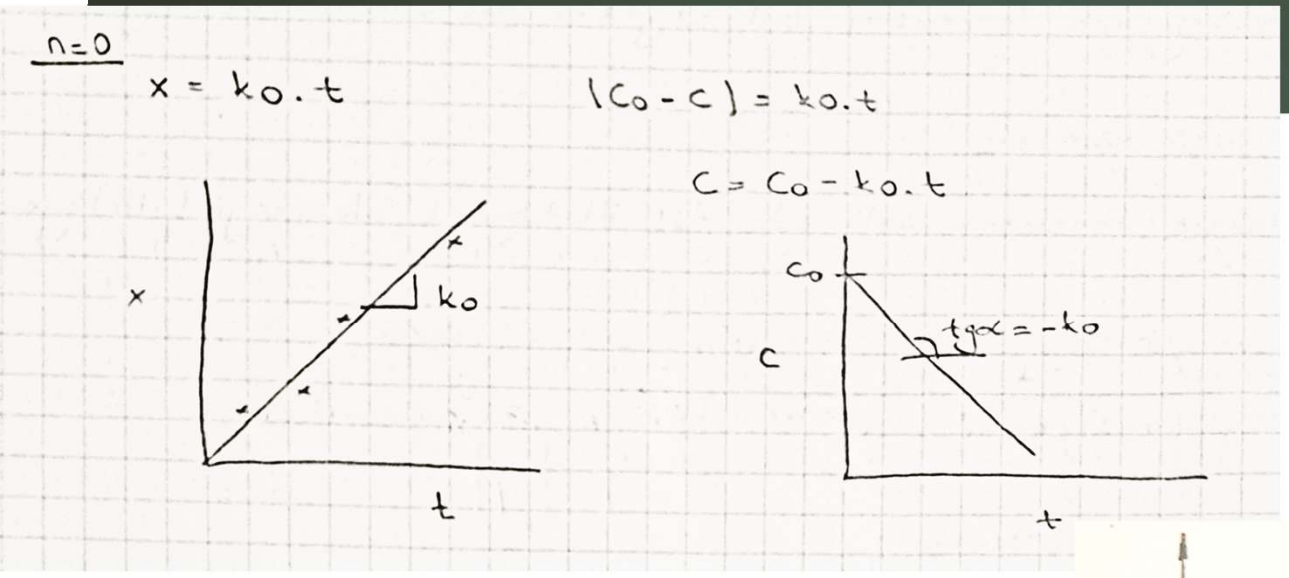
Derece	Hız Eşitliği: Diferansiyel denklem	Hız Eşitliği: İntegre edilmiş denklem	Hız Sabiti Birimi
0	$\frac{dx}{dt} = k$	$k = \frac{x}{t}$	mol/litre/sn
1/2	$\frac{dx}{dt} = k(C_0 - x)^{1/2}$	$k = \frac{2}{t}\left[C_0^{1/2} - x)^{1/2}\right]$	$mol^{1/2}/litre^{1/2}/sn$
1	$\frac{dx}{dt} = k(C_0 - x)$	$k = \frac{2.303}{t} \log \frac{C_0}{(C_0 - x)}$	1/ sn
3/2	$\frac{dx}{dt} = k(C_0 - x)^{3/2}$	$k = \frac{2}{t}\left[\frac{1}{(C_0 - x)^{1/2}} - \frac{1}{C_0^{1/2}}\right]$	$litre^{1/2}/mol^{1/2}/sn$
2	$\frac{dx}{dt} = k(C_0 - x)^2$	$k = \frac{1}{t}\left[\frac{x}{C_0(C_0 - x)}\right]$	litre/mol/sn
3	$\frac{dx}{dt} = k(C_0 - x)^3$	$k = \frac{1}{2t}\left[\frac{2C_0 x - x^2}{C_0^2 (C_0 - x)^2}\right]$	$litre^2/mol^2/sn$

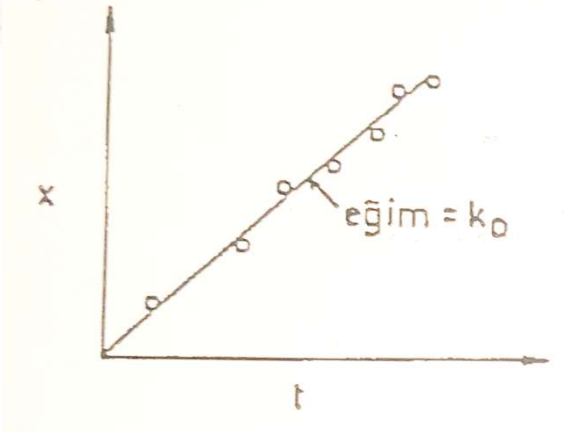
Linearized Coordinate / Graphic Method

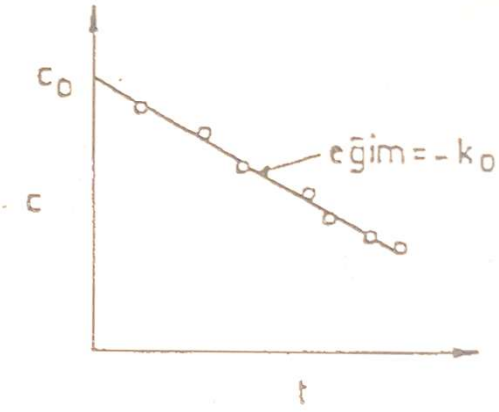
In this method, coordinate systems are choosen to show the experimental data as a line in the graph.

Zeroth order	$x = kt$	or	$C = C_0 - kt$
First oder	$ln\frac{a}{a-x} - t$	or	$lnC - t$
Second order			$\frac{1}{C} - t$
n^{th} order			$\frac{1}{C^{n-1}} - t$

Linearized Coordinate / Graphic Method







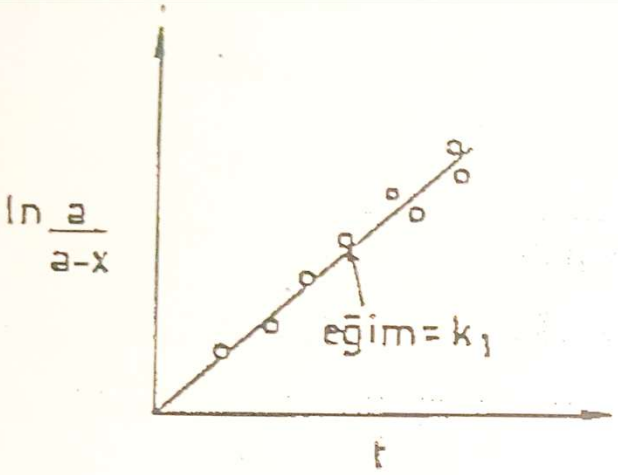
Linearized Coordinate / Graphic Method

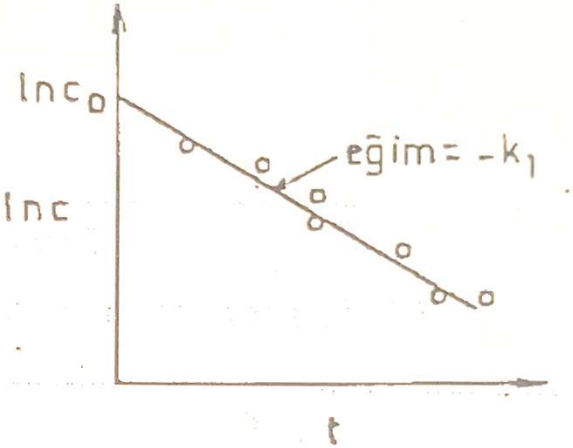
$n = 1$

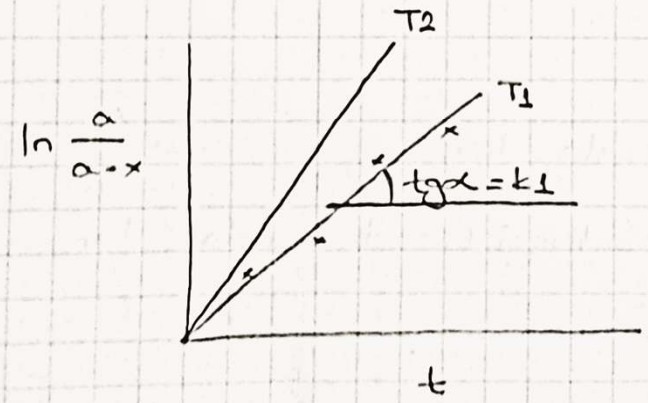
$$k_1 = \frac{1}{t} \ln \frac{a}{a-x}$$

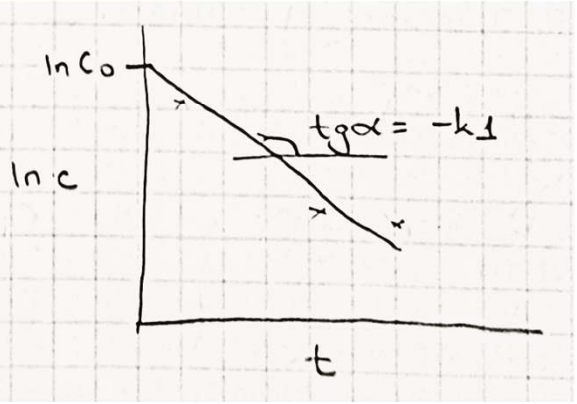
$$k_1 = \frac{1}{t} \ln \frac{C_0}{C}$$

$$k_1 . t = \ln C_0 - \ln C$$









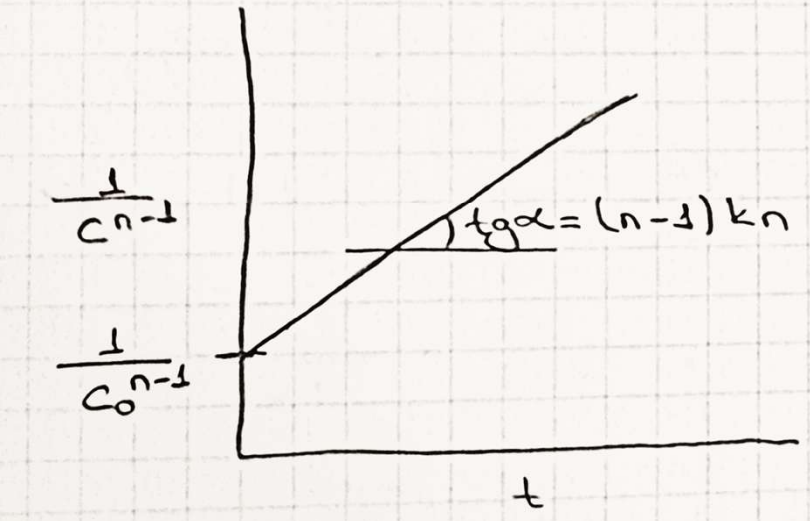
Linearized Coordinate / Graphic Method

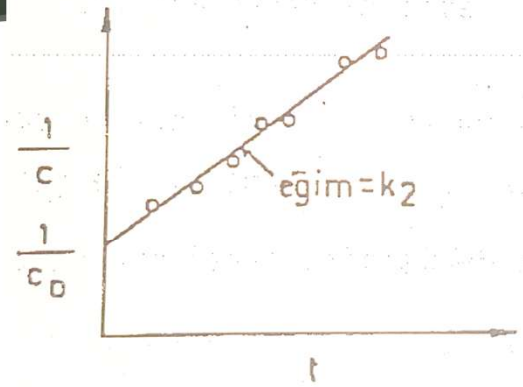
$n = n$

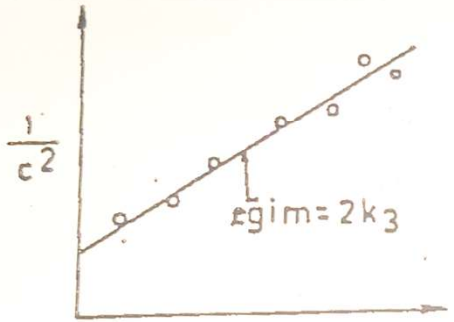
$$k_n = \frac{1}{(n-1)\,t}\left[\frac{1}{(a-x)^{n-1}} - \frac{1}{a^{n-1}}\right]$$

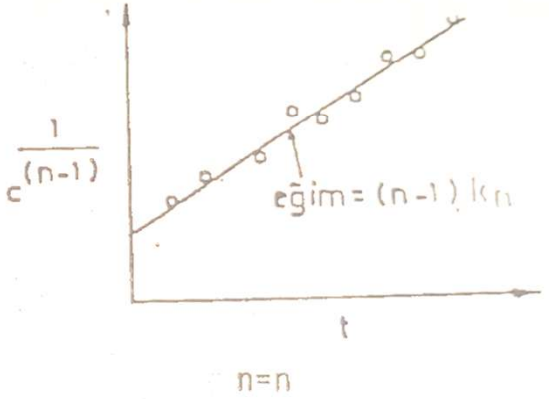
C^{n-1} $\qquad$ C_0^{n-1}

$$(n-1)\,k_n \cdot t = \dots$$









Half Life Method – Reaction Rate

The half life, $t_{1/2}$, of a substance is defined as the time it takes for the concentration of the substance to fall to half of its initial value.

It only makes sense to define a half life for a substance not present in excess at the start of the reaction.

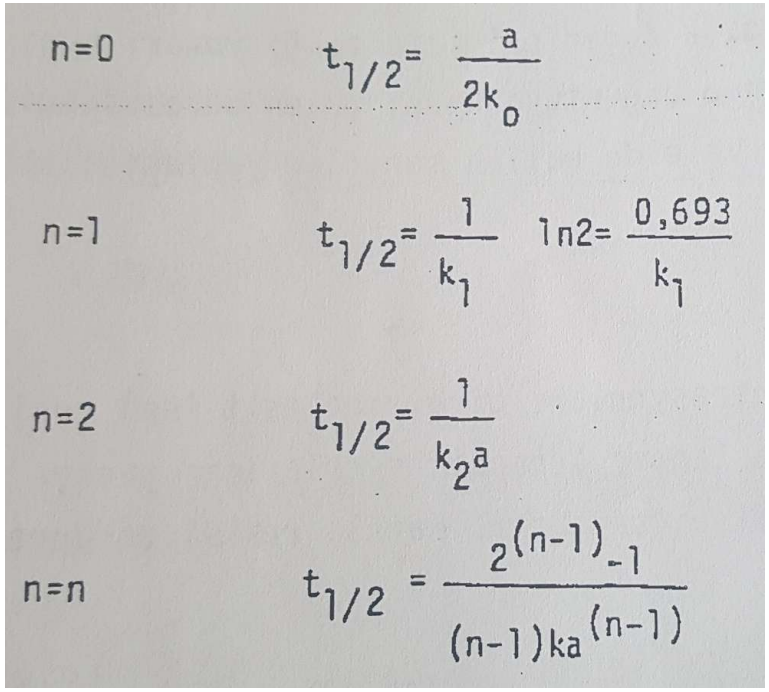
We can obtain equations for the half lives for reactions of various orders by substituting the values

$t = t_{1/2}$ and $[A] = ½ [A]_0$ (or a = a/2) into the integrated rate laws.

Zeroth order reaction	$t_{1/2} = \frac{[A]_0}{2k}$
First order reaction	$t_{1/2} = \frac{\ln 2}{k}$
Second order reaction	$t_{1/2} = \frac{1}{k[A]_0}$

Yarılanma Ömrü Yöntemi

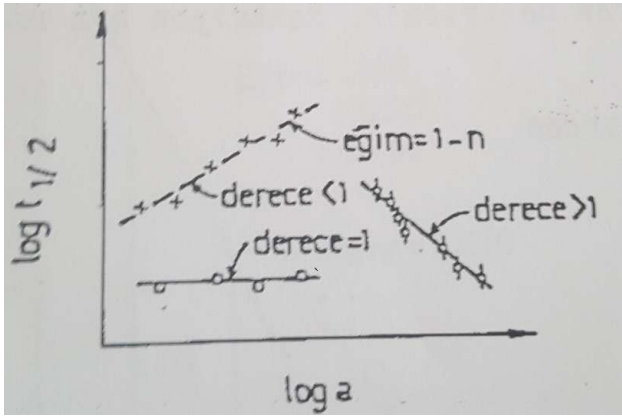
Bu yöntemde, reaktanların konsantrasyonları biribirine eşit ise, çeşitli reaksiyonlarda reaktanların başlangıç konsantrasyonlarının « **a** » yarıya düşmesi için gerekli yarılanma süreleri hesaplanır.



$$t_{1/2} = \frac{2^{n-1} - 1}{(n-1)ka^{(n-1)}}$$

$$log t_{1/2} = sabit - (n-1)\log a$$

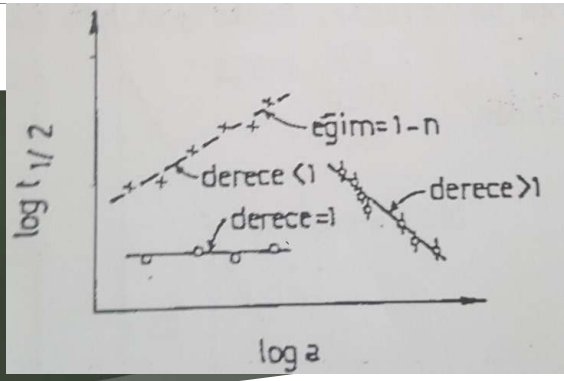
Eğer reaksiyonun yarılanma ömrü farklı başlangıç konsantrasyonları için biliniyorsa grafik ya da hesaplama yöntemi ile reaksiyon derecesi hesaplanabilir.



Example

$$log t_{1/2} = cst - (n-1)\log a$$

$$t_{1/2} = \frac{1}{k_2 a}$$



For A+B=C reaction, calculate the reaction order and the rate constant.

Experiment	Initial Concentration		Half-life Time (h)
	A	B	
1	2	2	2.5
2	5	5	1.0

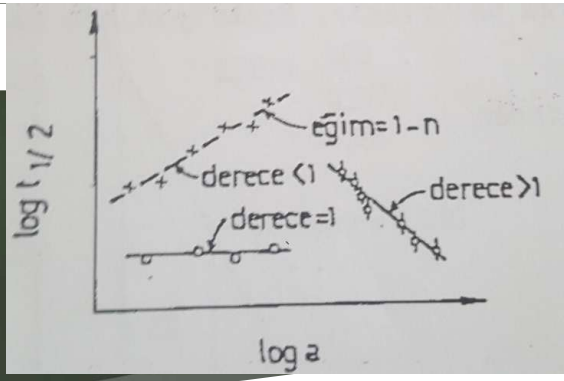
$$n = 1 + \frac{log t_{1/2\,1} - log t_{1/2\,2}}{log a_2 - log a_1} \qquad n = 1 + \frac{log 2.5 - log 1}{log 5 - log 2} \qquad n = 1 + \frac{log 2.5}{log 2.5} = 2$$

$$k = \frac{1}{a \times t_{1/2}} = \frac{1}{2 \times 2.5} = 0.2 \text{ liter/mol/hour}$$

Example

$$log t_{1/2} = cst - (n-1)\log a$$

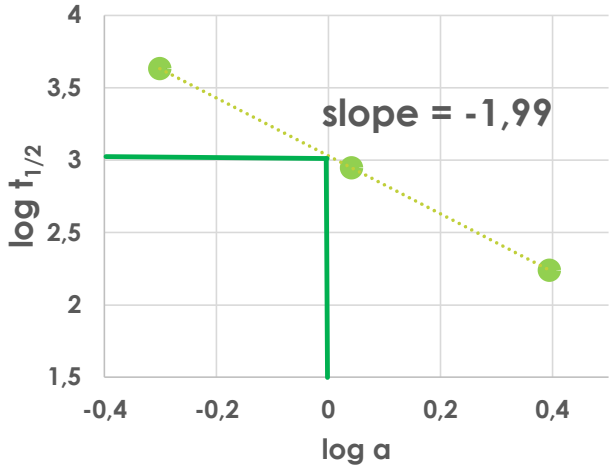
$$t_{1/2} = \frac{(2^{n-1}-1)}{(n-1)k_n a^{(n-1)}}$$



For a dissolution reaction, calculate the reaction order and rate constant.

Initial Concentration (mol/l)	0.5	1.1	2.48
Half-lite time (sec)	4280	885	174

log a	-0.3010	0.0414	0.3945
log $t_{1/2}$	3.6314	2.9464	2.2405



$$slope = -(n-1) = -1.99$$

$$n = 2.99 \cong 3$$

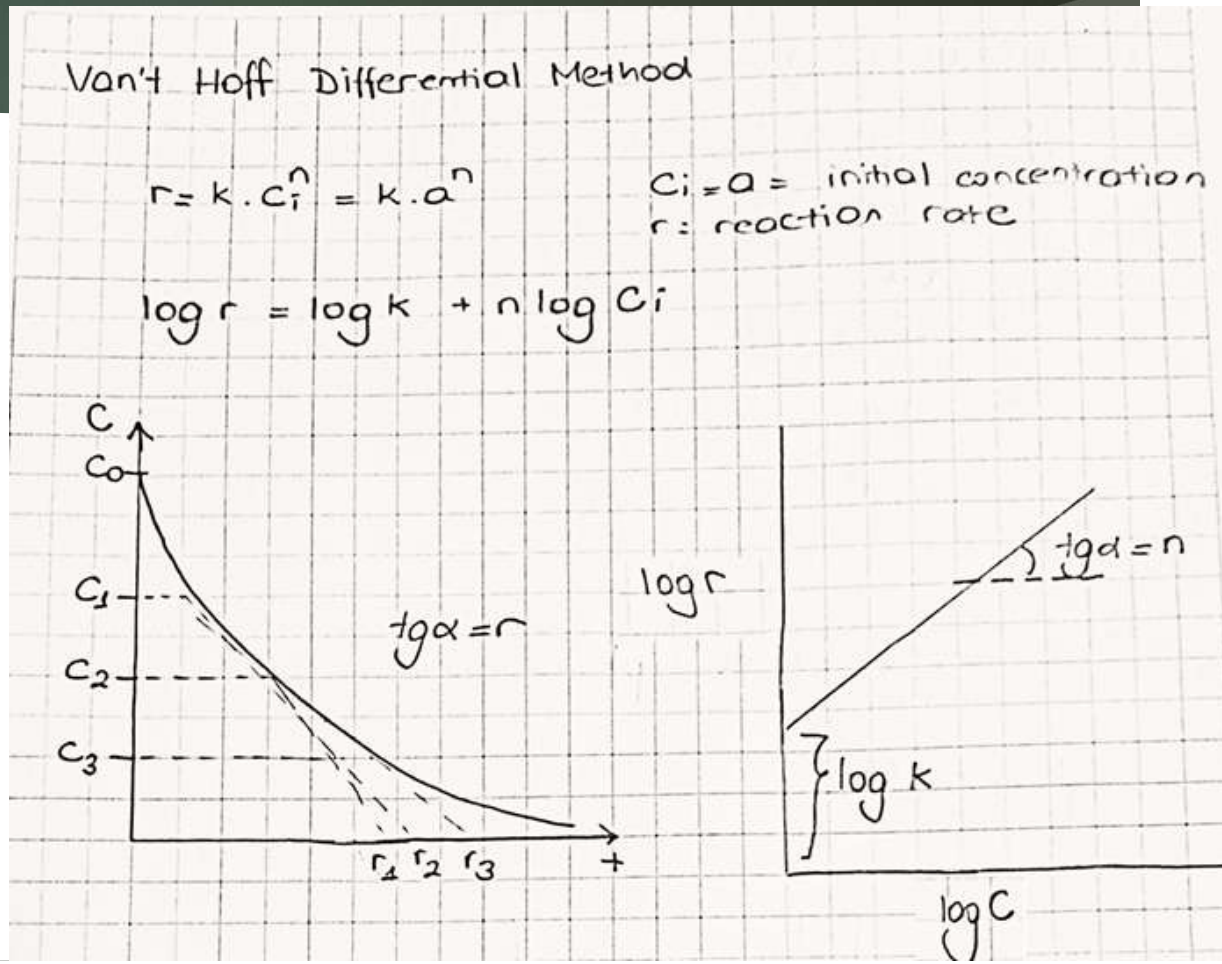
for log a = 0

$$log t_{1/2} = log \frac{2^{3-1}-1}{(3-1)k} = log \frac{3}{2k_3} = 3.03$$

$$k_3 = 1.399\times10^{-3} \text{ liter}^2/\text{mol}^2/\text{sec}$$

Van't Hoff Differential Method

In this method one utilizes no concentrations at given time but rates of reactions at given time.
It may be employed when reactions are not very fast. Most frequently it is utilized for initial rates.



Van't Hoff Differential Method

In two experiments performed with differential initial concentration:

$$v_1 = kc_1^n \quad (1)$$

$$v_2 = kc_2^n \quad (2)$$

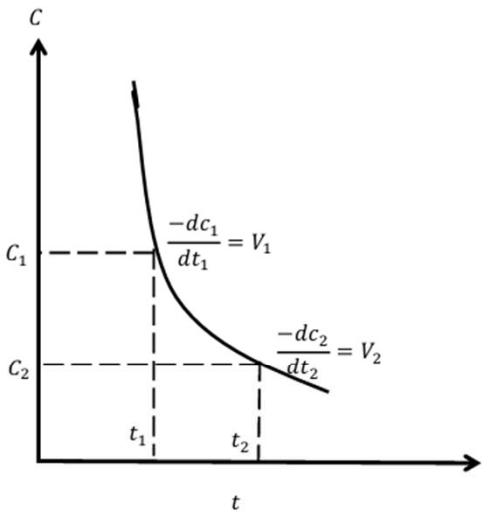
By use the log:

$$\log v_1 = logK + n \log C_1 \quad (3)$$

$$\log v_2 = logK + n \log C_2 \quad (4)$$

Subtracting 4 from 3:

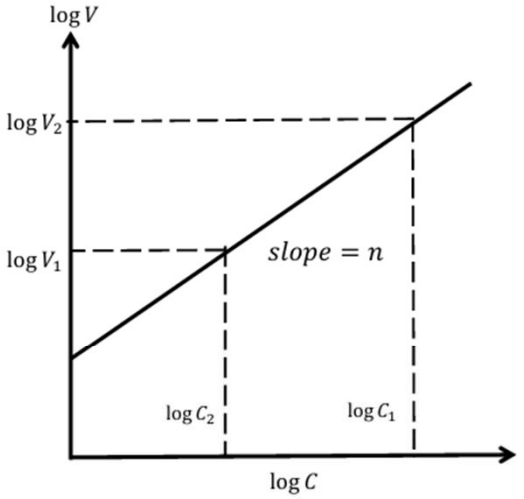
$$n = \frac{\log v_1 - \log v_2}{\log c_1 - \log c_2}$$



he equation 1,2:

$$v_1 = kc_1^n \quad (1)$$

$$v_2 = kc_2^n \quad (2)$$



The equation 3,4:

$$\log v_1 = logK + n \log C_1 \quad (3)$$

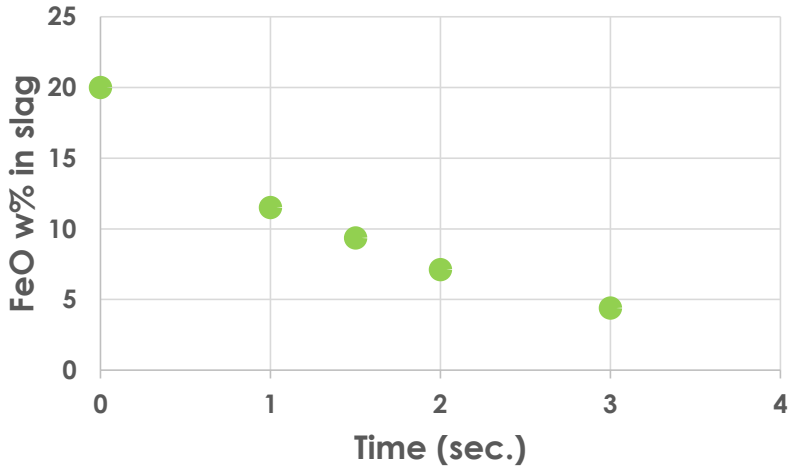
$$\log v_2 = logK + n \log C_2 \quad (4)$$

Example

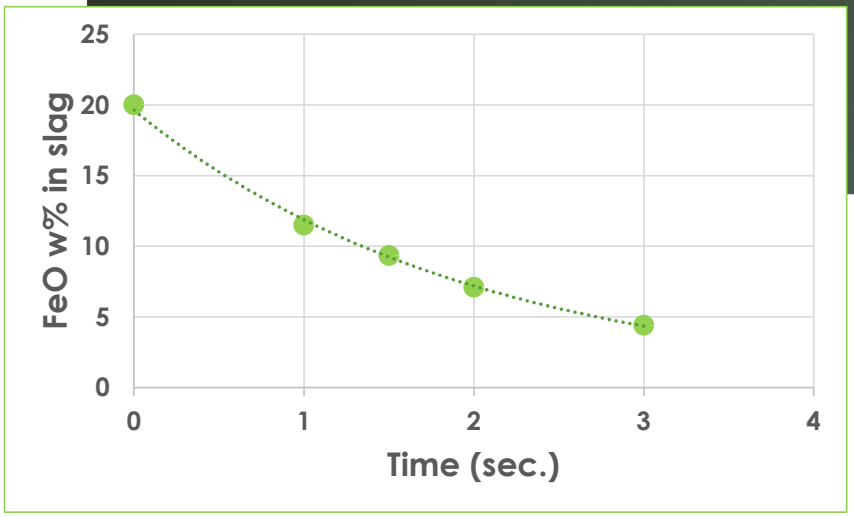
The reduction kinetics at a certain temperature between the C in liquid iron and FeO in slag is observed and obtained data is given below.

According to the table, determine the order of reaction by using Van't Hoff differential method and calculate the reaction rate constant (k).

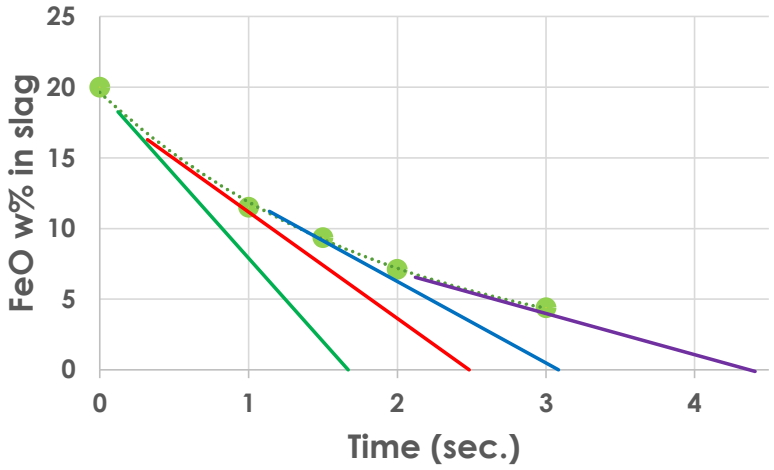
FeO in slag(w%)	20.00	11.50	9.35	7.10	4.40
Time (second)	0	1.0	1.5	2.0	3.0

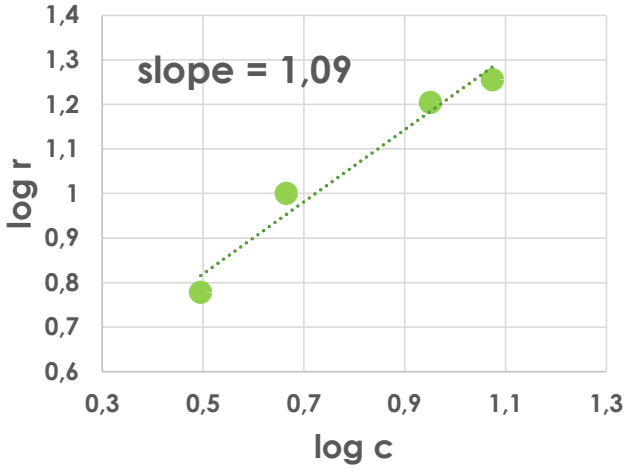


$$\log r = \log k + n \log C$$



Tangent #	FeO in slag (w%) - C	log C	Slope of tangent	Reduction rate (r)	log r
1	18	1.2553	-11.88	11.88	1.0749
2	16	1.2041	-8.94	8.94	0.9513
3	10	1.000	-4.63	4.63	0.6656
4	6	0.7782	-3.13	3.13	0.4955





$$r = kC^n$$

$$\log r = \log k + n \log C$$

$$\text{at } \log C = 0$$

$$\log k = -0{,}34$$

$$k = 0{,}4571 \text{ l/min}$$

A solid-state reduction reaction of nickel oxide with hydrogen gas is studied at 800 °C:

$$\mathrm{NiO(s)} + \mathrm{H_2}(g) \rightarrow \mathrm{Ni(s)} + \mathrm{H_2O(g)}$$

The following experimental data were obtained for different hydrogen partial pressures at constant temperature:

P_{H_2} (atm)	Reaction rate, r (mol NiO reduced / s·m^2)
0.2	1.8×10^{-6}
0.4	3.5×10^{-6}
0.8	7.0×10^{-6}

1. Using the **Van't Hoff Differential Method**, determine the **reaction order (n)** with respect to hydrogen.
2. Write the corresponding rate law expression.
3. Discuss briefly what a first-order dependence on H_2 would imply about the rate-controlling step in this solid–gas reaction.

$$\log r = \log k + n \log P_{H_2}$$

P_{H_2} (atm)	r (mol\cdotps$^{-1}\cdot$m^{-2})
0.2	1.8×10^{-6}
0.4	3.5×10^{-6}
0.8	7.0×10^{-6}

$$n = \frac{\ln(r_2/r_1)}{\ln(P_2/P_1)}$$

$P = 0.2 \to 0.4$:

$$n = \frac{\ln(3.5 \times 10^{-6}/1.8 \times 10^{-6})}{\ln(0.4/0.2)} \approx 0.959$$

$P = 0.4 \to 0.8$:

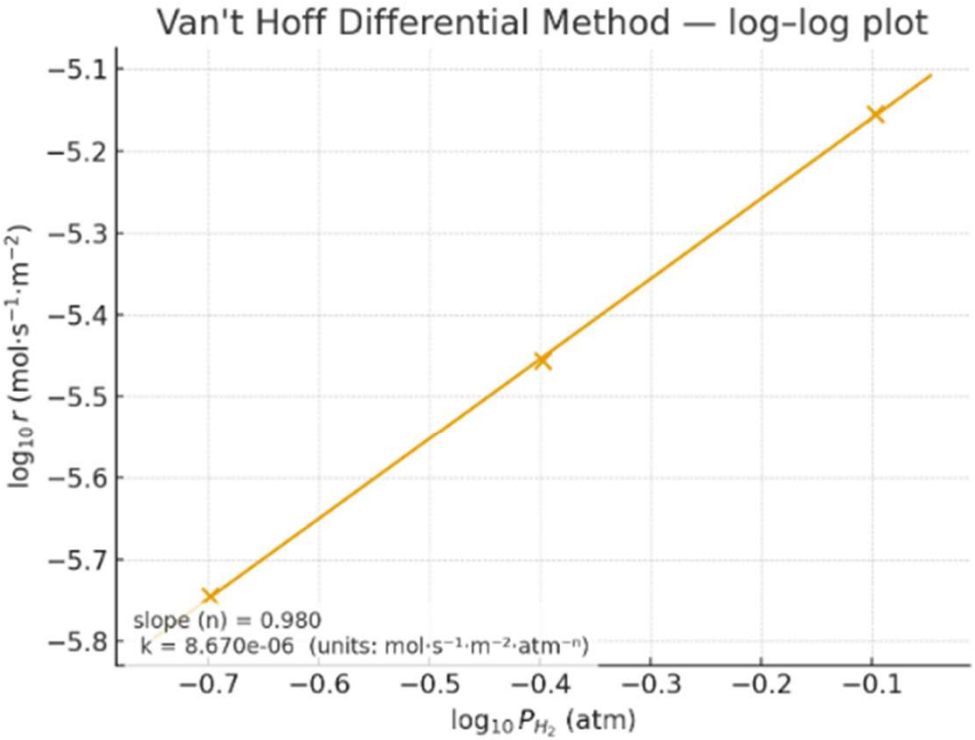
$$n = \frac{\ln(7.0 \times 10^{-6}/3.5 \times 10^{-6})}{\ln(0.8/0.4)} = 1.00$$

$P = 0.2 \to 0.8$:

$$n = \frac{\ln(7.0 \times 10^{-6}/1.8 \times 10^{-6})}{\ln(0.8/0.2)} \approx 0.9797$$

$$k_1 = 1.8 \times 10^{-6}/0.2 = 9.0 \times 10^{-6}$$
$$k_2 = 3.5 \times 10^{-6}/0.4 = 8.75 \times 10^{-6}$$
$$k_3 = 7.0 \times 10^{-6}/0.8 = 8.75 \times 10^{-6}$$

$$k \approx 8.83 \times 10^{-6}\ \text{mol\cdotps}^{-1}\cdot\text{m}^{-2}\cdot\text{atm}^{-1}$$



According to the molecularity

The overall, balanced chemical equation does not tell us how a reaction actually takes place. It describes the products and the reactants, and the relative ratios in which they are consumed and produced. In many cases, it represents the sum of a series of more simple reactions, that are called the **elementary steps**. **They represent the progress of the overall reaction at a molecular level.** The sequence of elementary steps that lead to product formation is called the **reaction mechanism.** Each elementary step will have a **molecularity**. The molecularity is the number of molecules that react in an elementary process. There are 3 possible molecularities:

Molecularity	Elementary step	rate law
*Uni*molecular	$A \rightarrow$ product	$k[A]$
*Bi*molecular	$A + B \rightarrow$ product	$k[A][B]$
	$A + A \rightarrow$ product	$k[A]^2$
*Ter*molecular	$A + B + C \rightarrow$ product	$k[A][B][C]$
	$A + A + A \rightarrow$ product	$k[A]^3$
	$A + A + B \rightarrow$ product	$k[A]^2[B]$

It is important to distinguish between molecularity and order. The molecularity is the number of molecules that actually react in an elementary process, and must be a small integer, either 1, 2 or rarely, 3. In contrast, the order is the sum of the exponents in the rate law for the overall reaction. It can only be determined experimentally and can be 0, fractional or integral.

The difference between Order of reaction and Molecularity explained below

Sl. No.	Molecularity	Order of Reaction
1	It is the number of reacting species undergoing collisions the reactions.	It is the sum of the power of the conc. terms in the rate law expression.
2	It is a theoretic value.	It is determined experimentally
3	It has positive integral value 1, 2,3…n.	It can have negative & fractional value also.
4	It does not tell us anything about the mechanism of the reaction.	It tells us about some clue about mechanism of the reaction.
5	It does not change with change in temperature and pressure.	It may change with change in temperature and pressure.
6	It is the property of only elementary reactions and has no meaning for complex reactions.	It is the property of both elementary and complex reaction.

Reversible Reactions Kinetics

$$A \underset{2}{\overset{1}{\leftrightarrow}} B$$

$$\frac{dx}{dt} = \left(\frac{dx}{dt}\right)_{right} - \left(\frac{dx}{dt}\right)_{left}$$

$$k\ (rate\ constant) = \frac{k_1}{k_2}$$

$$\frac{dx}{dt} = k_1(a - x) - k_2 x$$

a: initial concentration of A
x: concentration of B at time t

The net rate of the reaction is directly proportional to how far the system is from equilibrium.

$$\boxed{\frac{dx}{dt} = (k_1 + k_2)(x_{eq} - x)}$$

For the equilibrium condition

x_{eq}: product or reacted part of reactant at equilibrium condition

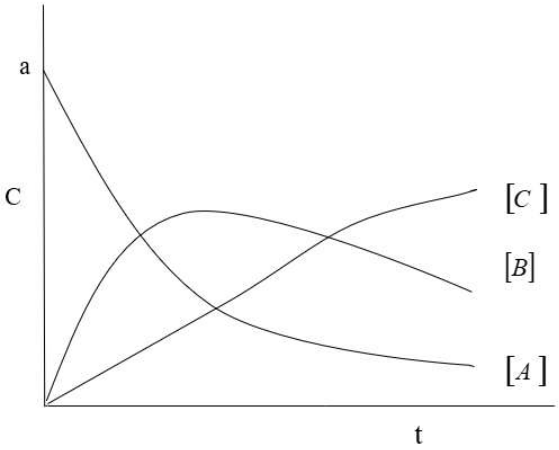
$$k_1(a - x_{eq}) = k_2 x_{eq}$$

$$\frac{dx}{dt} = \frac{k_1 a}{x_{eq}}(x_{eq} - x)$$

$$\frac{k_1 a}{x_{eq}} = \frac{1}{t} ln \frac{x_{eq}}{(x_{eq} - x)}$$

$$\frac{k_1 a}{x_{eq}} = k_1 + k_2$$

Consecutive (Serial) Reactions Kinetics



$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

$$-\frac{d[A]}{dt} = k_1\,[A]$$

$$\ln[A] = -k_1 t + \ln a$$

$$[A] = ae^{-k_1 t}$$

$$\frac{d[B]}{dt} = k_1[A] - k_2[B]$$

$$\frac{d[B]}{dt} = k_1 ae^{-k_1 t} - k_2[B]$$

$$[B] = \frac{a^{k_1}}{k_2 - k_1}[e^{-k_1 t} - e^{-k_2 t}]$$

$$[A] + [B] + [C] = a$$

$$[C] = a - ([A] + [B])$$

$$[C] = a[1 - \frac{k_2}{k_2 - k_1} e^{-k_1 t} + \frac{k_1}{k_2 - k_1} e^{-k_2 t}]$$

If $k_2 >> k_1$

$$[C] = a[1 - e^{-k_1 t}]$$

If $k_1 >> k_2$

$$[C] = a[1 - e^{-k_2 t}]$$

Parallel Reactions Kinetics

$$A \xrightarrow{1} B$$
$$A \xrightarrow{2} C$$

$$r_A = -\frac{dC_A}{dt} = k_1 C_A + k_2 C_A = (k_1 + k_2) C_A$$

$$r_B = \frac{dC_B}{dt} = k_1 C_A$$

$$r_C = \frac{dC_C}{dt} = k_2 C_A$$

$$\boxed{\frac{r_B}{r_C} = \frac{k_1}{k_2}}$$

Effect of temperature on reaction rate

The reason for the increase in reaction rate with increasing temperature

- Collisions of molecules???
  - Not the only cause

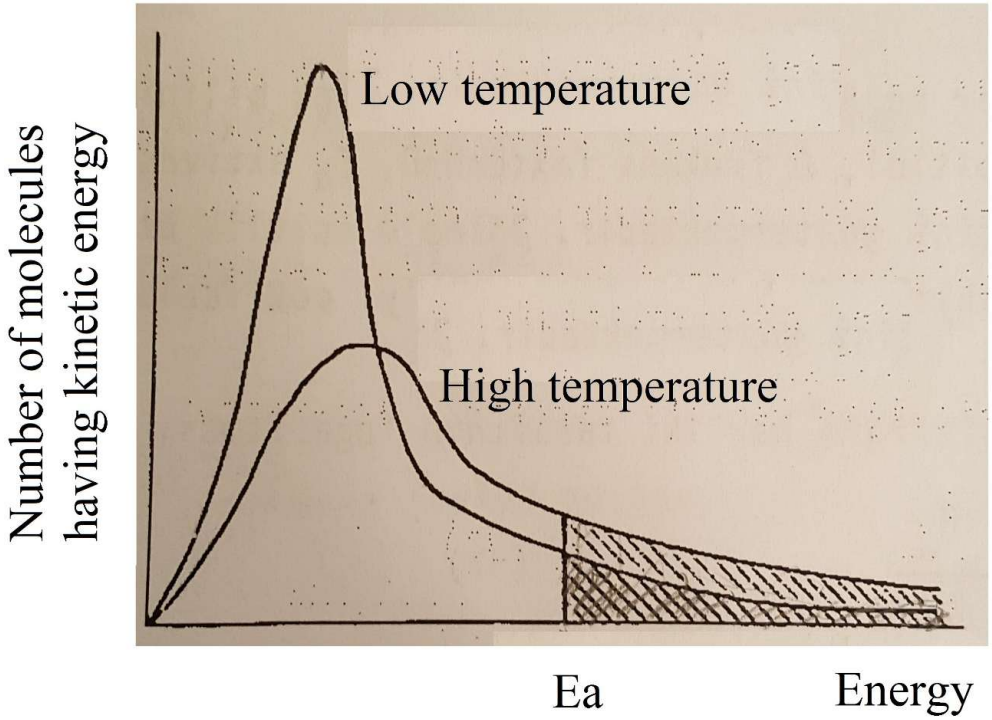
Collision of molecules is necessary but not sufficient for a reaction takes place
For a reaction, molecules should have a certain energy namely **activation energy**

Activation energy = Initial energy of system (before reaction) – Energy required for the reaction

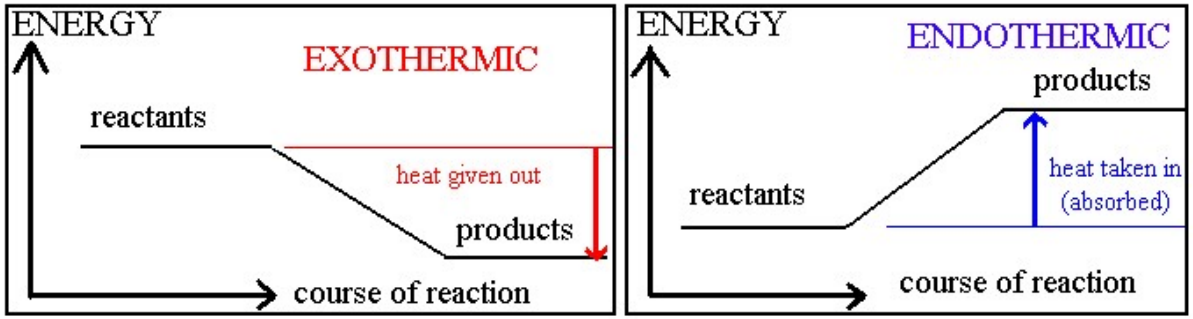
Molecules having activation energy are named as activated molecules

Number of the activated molecules is another cause of the increase in reaction rate with changing temperature

Boltzmann Distribution Law



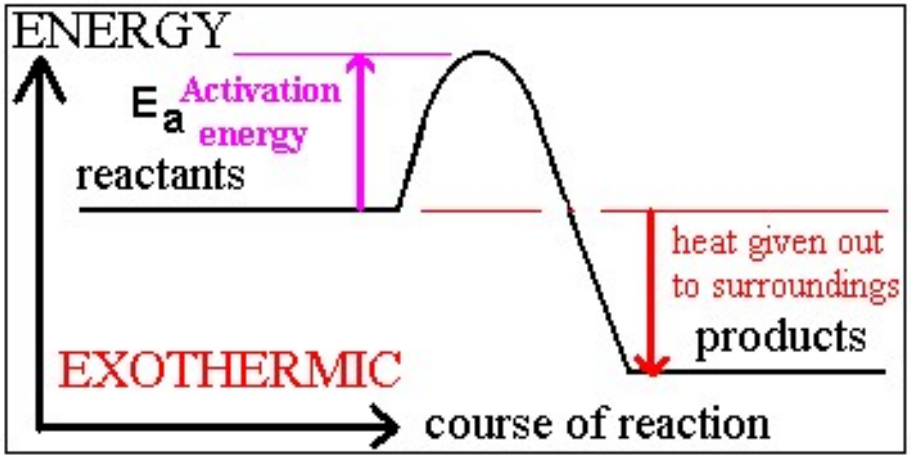
Effect of temperature on reaction rate

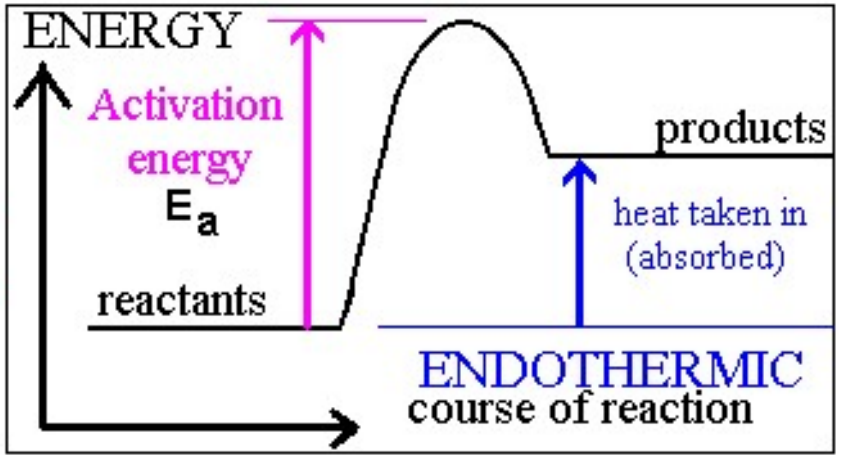


- When **gases or liquids are heated the particles gain kinetic energy** and move faster increasing the chance of collision between reactant.
- **However** this is not the main reason for the increased reaction speed on increasing the temperature of reactant molecules **because most molecular collisions do not result in chemical change**.
- **Before any change takes place on collision, the colliding molecules must have a minimum kinetic energy** called the **activation energy** ($\mathbf{E_a}$).
- Do not confuse activation energy with the overall energy change, the overall energy absorbed by the system (endothermic) or given out to the surroundings (exothermic).
- It does not matter whether the reaction is an exothermic or an endothermic energy change.
- **Higher temperature molecules in gases and liquids have a greater average kinetic energy and so a greater proportion of them will then have the required activation energy to react on collision.**
- **The increased chance of higher energy collisions greatly increases the speed of the reaction because it greatly increases the chance of a fruitful collision forming the reaction products** by bonds being broken in the reactants and new bonds formed in the reaction products**.**

Effect of temperature on reaction rate

Energy level diagrams for exothermic and endothermic reactions showing the activation energy (E_a) 'hump' which must be overcome before a chemical reaction can take place

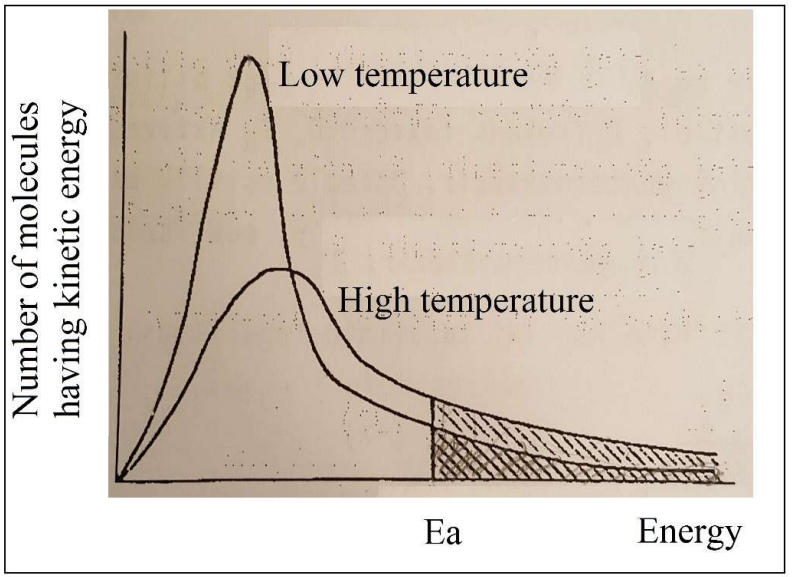




Boltzmann Distribution Law

Rate of the number of molecules having a certain activation energy (E_a) to the total number of molecules

$$\frac{\Delta N}{N} = e^{-E_a/RT}$$



At higher temperatures

- number of molecules having E_a increases
- reaction rate increases

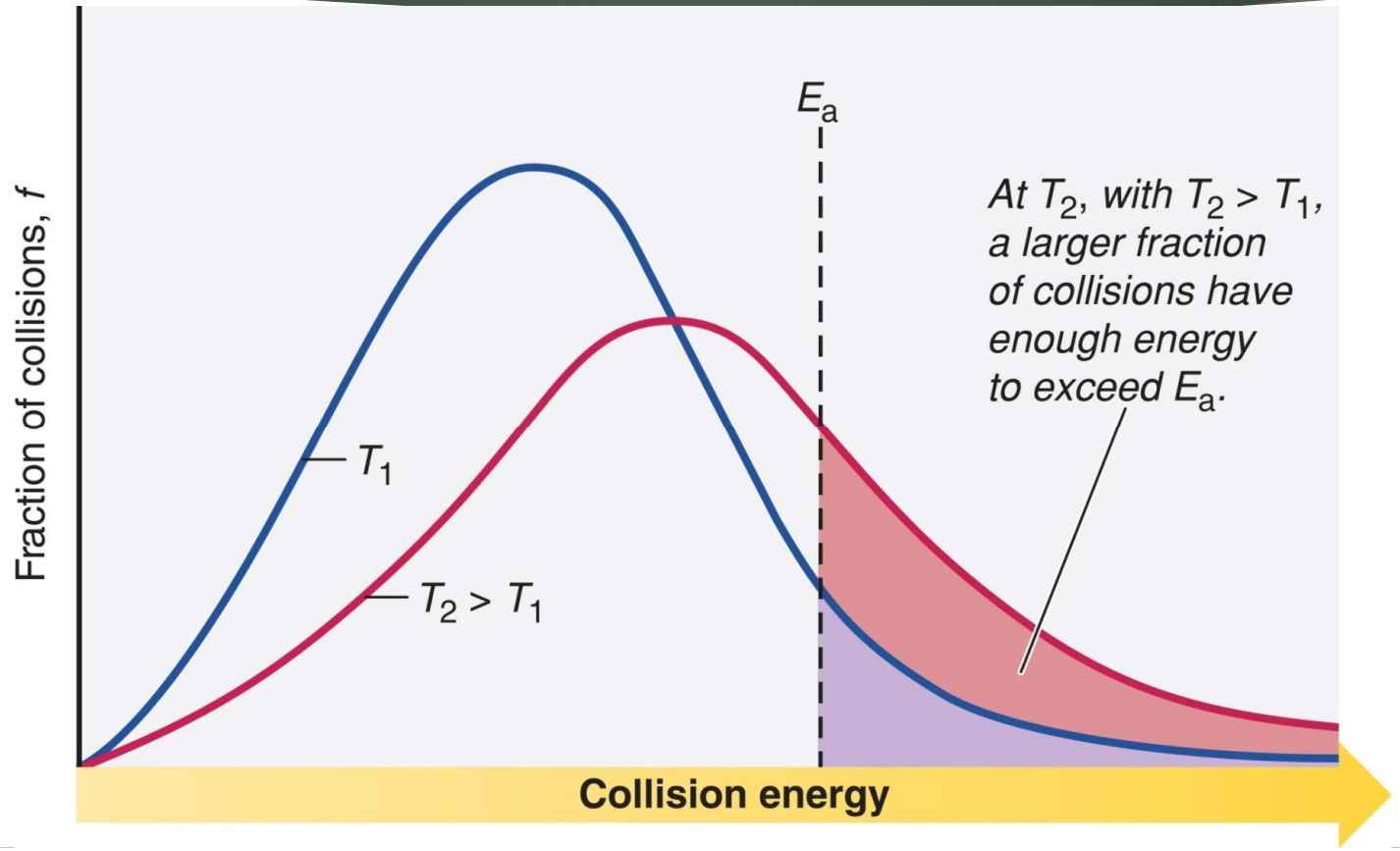
$$k = A \times e^{-E_a/RT}$$ **Arrhenius Equation**

k: is greatly affected by temperature

A: frequency factor
E_a : Activation energy
R: Gas constant
T: temperature
k: rate constant

$$lnk = lnA - \frac{E_a}{RT}$$

The effect of temperature on the distribution of collision energies

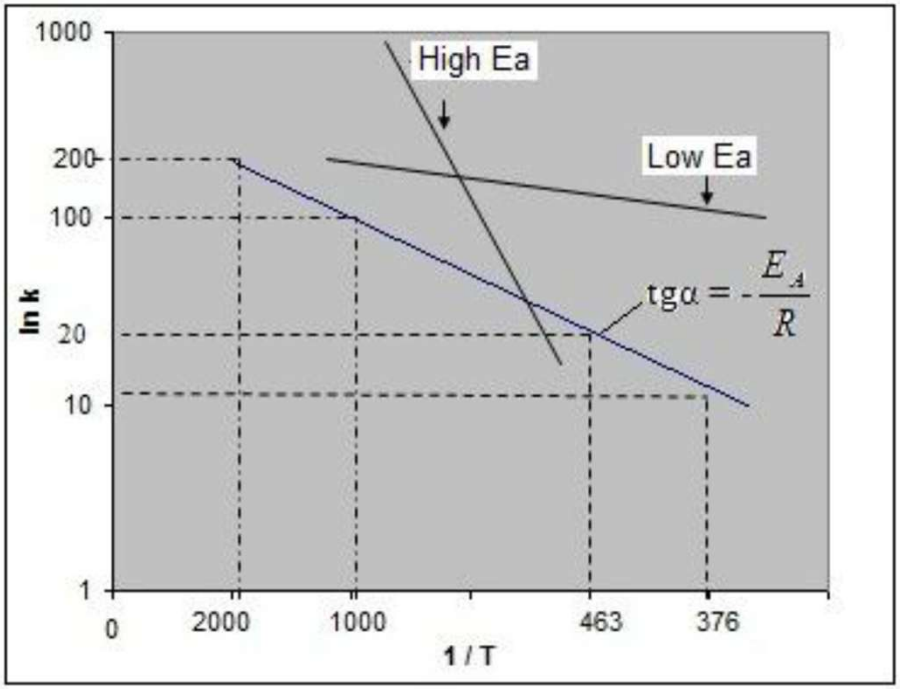


Activation Energy depending on temperature

- One of the most important parameters affecting the rate of chemical reactions is temperature.
- In order to calculate the activation energy from Arrhenius Equation rate constants (*k*) should be determined at least for two different temperatures.
- Reaction rate is exponentially dependent on temperature. Therefore, a small change in the temperature increases the rate constant dramatically.
- By taking the logarithm of both sides in the Arrhenius equation, the ln *k* – 1/*T* change can be plotted to calculate the activation energy.
- Slope of the obtained curve is equal to ($-E_a$ / R) value.

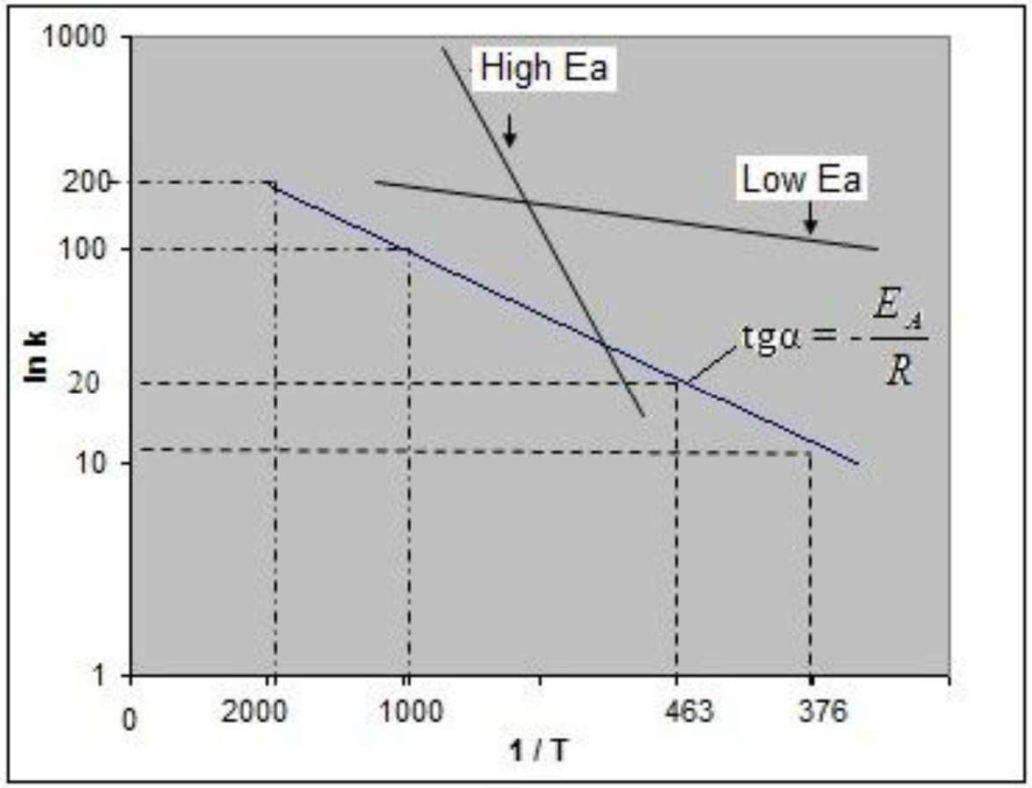
$$k = A \times e^{-E_a/RT}$$

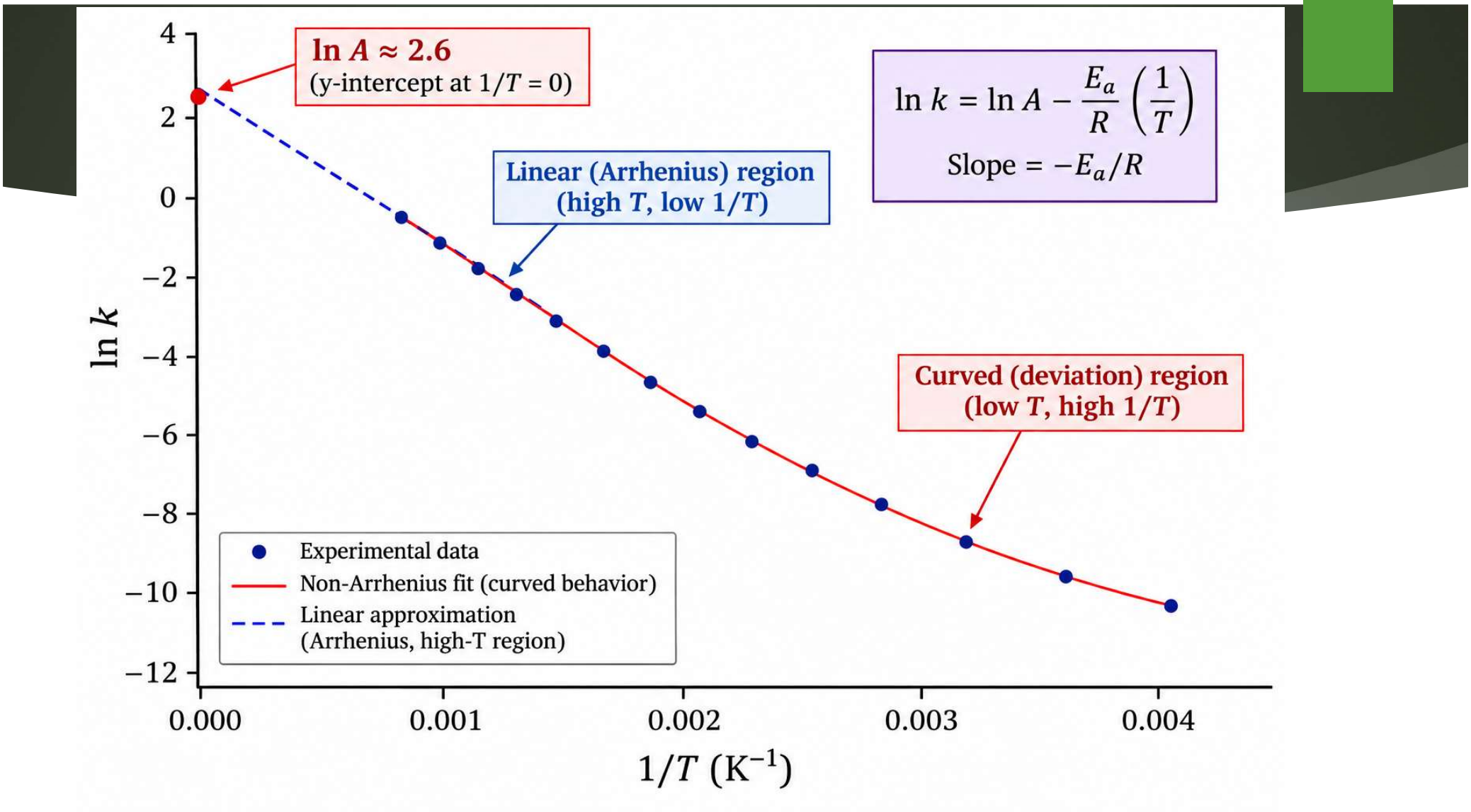
$$lnk = lnA - \frac{E_a}{RT}$$



Activation Energy depending on temperature

- The **higher is the slope** of the straight line the **higher is the activation energy**.
- Comparing with reactions which have low activation energies, **reactions with high activation energies are more dependent on temperature**. This means that they are more affected by the temperature changes.
- **Reactions realized at lower temperatures are more dependent on temperature** than the reactions realized at higher temperatures.
- **Pre-exponential factor** (prefactor) is not affected by the temperature dependence of the reaction so it **is not dependent on temperature.**



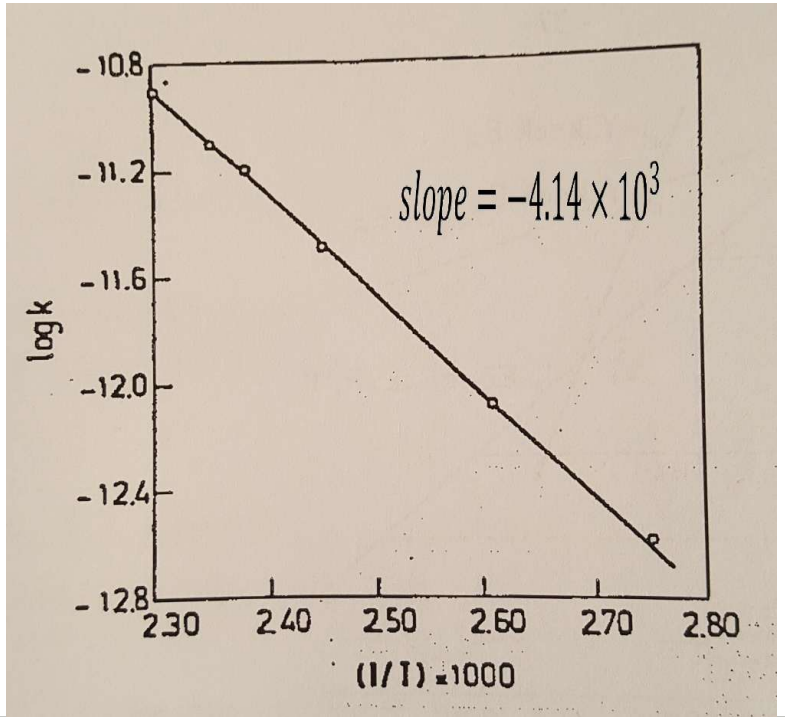

ln A ≈ 2.6
(y-intercept at 1/T = 0)
ln k = ln A − (Ea/R)(1/T)
Slope = −Ea/R
Linear (Arrhenius) region
(high T, low 1/T)
Curved (deviation) region
(low T, high 1/T)
Experimental data
Non-Arrhenius fit (curved behavior)
Linear approximation
(Arrhenius, high-T region)
ln k
1/T (K⁻¹)
4
2
0
−2
−4
−6
−8
−10
−12
0.000
0.001
0.002
0.003
0.004

Example

$$lnk = lnA - \frac{E_a}{RT}$$

At a constant oxygen pressure and different temperatures, reaction rate constants of the leach reaction of galen in an ammonium acetate solution are given in table below. Calculate the activation energy for the leach reaction.

log k mol²/cm⁴/min	-10.9	-11.1	-11.2	-11.5	-12.1	-12.6
$\frac{1}{T}\times 10^3$ K⁻¹	2.3	2.35	2.38	2.45	2.61	2.71



$$logk = logA - \frac{E_a}{19.147T}$$

$$tg\alpha = -\frac{E_a}{19.147} = -4.14\times 10^3$$

$$\boldsymbol{E_a = 79268\ J}$$

Example

Reaction rate constant of a reaction at 298 K is $k=5.03\times10^{-2}\ mol^{-1}s^{-1}$ and at 333 K $k=6.71\ mol^{-1}s^{-1}$.

- Calculate the activation energy.
- Calculate the reaction rate constant at 306 K.

$$logk = logA - \frac{E_a}{19.147T}$$

$$log5.03\times10^{-2} = logA - \frac{E_a}{19.147\times298}$$

$$log6.71 = logA - \frac{E_a}{19.147\times333}$$

$$log5.03\times10^{-2} - log6.71 = \frac{E_a}{19.147}\left(\frac{1}{298} - \frac{1}{333}\right)$$

$$E_a = 50.075\ kJ$$

Example

Reaction rate constant of a reaction at 298 K is $k=5.03\times10^{-2}\ mol^{-1}s^{-1}$ and at 333 K $k=6.71\ mol^{-1}s^{-1}$.

- Calculate the activation energy.
- Calculate the reaction rate constant at 306 K.

$$log6.71 = logA - \frac{50075}{19.147\times333}$$

$$logk = logA - \frac{50075}{19.147\times306}$$

$$k = 0.36mol^{-1}s^{-1}$$